

INDIGENE

GROSS ANATOMY OF THE LOWER LIMB

ANA217 — Complete Exam-Ready Compendium

Osteology · Joints & Ligaments · Nerve Supply · Musculature · Vasculature · Master Hack Sheet

200 Level — Nursing Science, Federal University Oye-Ekiti (FUOYE)
INDIGENE Study Console · Built for full exam mastery

TABLE OF CONTENTS

1. Osteology of the Lower Limb

Hip Bone · Femur · Patella, Tibia & Fibula · Bones of the Foot + *Complement: Duverney fracture, Coxa vara/valga, Femoral fracture sites, Crus fracture*

2. Joints and Ligaments of the Lower Limb

Hip Joint · Knee Joint (incl. Bursae) · Ankle Joint · Ligaments of the Sole of the Foot + *Complement: Pipkin fracture*

3. Nerve Supply — Lumbosacral Plexus

Lumbar Plexus · Sacral Plexus · Motor Nerves · Full Cutaneous Nerve Map (Thigh, Leg, Foot) · Erb's-equivalent Clinical Correlations + *Complement: Sciatic Nerve composition & course*

4. Musculature of the Lower Limb

Gluteal Region · Thigh (Anterior/Medial/Posterior) · Leg (Anterior/Lateral/Posterior) · Intrinsic Foot Muscles

5. Vasculature of the Lower Limb

Femoral Artery & Branches · Profunda Femoris · Obturator Artery · Popliteal Artery & Genicular Anastomosis · Anterior/Posterior Tibial · Dorsalis Pedis & Plantar Arteries + *Complement: DVT, Varicose Veins, Vascular Grafting (GSV)*

6. Femoral Triangle & Popliteal Fossa

7. Master Hack Sheet

8. New Additions — Quick-Hack Addendum

Compressed exam hacks for every complement topic added in this edition

1. OSTEOLOGY OF THE LOWER LIMB

The bony skeleton of the lower limb comprises the **hip bone** (pelvic girdle), the **femur** (thigh), the **patella, tibia, and fibula** (leg), and the **tarsals, metatarsals, and phalanges** (foot).

Hip Bone (Os Coxae)

Alzheimer's Mode: Three bones fused into one, meeting like 3 roads converging at a roundabout — the acetabulum. Before puberty, ilium, ischium, and pubis are separate, joined by a Y-shaped cartilage.

Bone	Position	Acetabular contribution
Ilium	Superior, fan-shaped	Upper 2/5
Ischium	Posteroinferior	Posteroinferior 2/5
Pubis	Anteroinferior	Anterior 1/5

Ilium: iliac crest, ASIS/PSIS, greater sciatic notch. Ischium: ischial tuberosity (hamstring origin), ischial spine (pudendal nerve block landmark). Pubis: pubic symphysis.

EXAM HACK
Tuberosity = sit on it, hamstrings. Spine = smaller, superior, pudendal nerve block landmark. Don't swap them.

Femur

Alzheimer's Mode: The longest, strongest, heaviest bone in the body — a crowbar angled inward so the knees sit closer together than the hips (angle of inclination, greater in women due to a wider pelvis).

Proximal End

- **Head** — spherical, two-thirds covered by hyaline cartilage; articulates with the acetabulum to form the hip joint. Bears the **fovea (fovea capitis)** on its upper surface — a depression holding the **ligamentum teres (foveal ligament)**, which anchors the head to the pelvis. The **foveal (ligamentum teres) artery** travels within this ligament, contributing to femoral head blood supply.
- **Neck** — a bony stalk directed superomedially from the shaft; the **most fracture-prone part of the femur** due to its small size and inherent structural weakness.
- **Greater trochanter** — large, irregular, lateral prominence; major attachment site for the gluteal muscles.
- **Lesser trochanter** — small, rounded prominence, posteromedial, ~3cm below the greater trochanter's lower edge; attachment for iliopsoas.
- **Quadratus tubercle** — a small bony prominence (smaller than the lesser trochanter) on the posterior femoral surface, just inferior to the greater trochanter, medial to the intertrochanteric crest; attachment site specifically for **quadratus femoris**.

ADDED DETAIL
The **quadratus tubercle** and the **foveal (ligamentum teres) artery** are two specific landmarks worth naming precisely — the tubercle is quadratus femoris's dedicated attachment point, and the foveal artery is the named vessel (a branch ultimately from the obturator artery) running inside the ligament of the head of femur to help supply the femoral head, especially important in childhood.

Shaft

- **Gluteal tuberosity** — longitudinal posterior ridge, proximal continuation of linea aspera; gluteus maximus attaches here.
- **Linea aspera** — sharp longitudinal roughening on the posterior shaft, most prominent midway along the shaft, diverging at both proximal and distal ends; attachment for multiple lower limb muscles (adductors, vastus lateralis/medialis, short head of biceps femoris).
- **Pectineal line** — runs on the medial/posterior shaft from the inferior edge of the lesser trochanter to the medial (proximal) diverging margin of linea aspera; attachment for pectineus.

Distal End

- **Epicondyles** (medial + lateral) — bony protuberances for muscle/ligament attachment.
- **Medial and lateral supracondylar lines** — formed by the distal divergence of the linea aspera, providing further muscle attachment (adductor magnus inserts at the adductor tubercle above the medial epicondyle).
- **Condyles** (medial + lateral) — fully cartilage-covered, form the knee joint. The anterior surface of the distal femur is cartilage-covered too, called the **patellar surface**, articulating with the patella.
- **Intercondylar fossa** — deep posterior hollow between the condyles; attachment site for the cruciate ligaments.

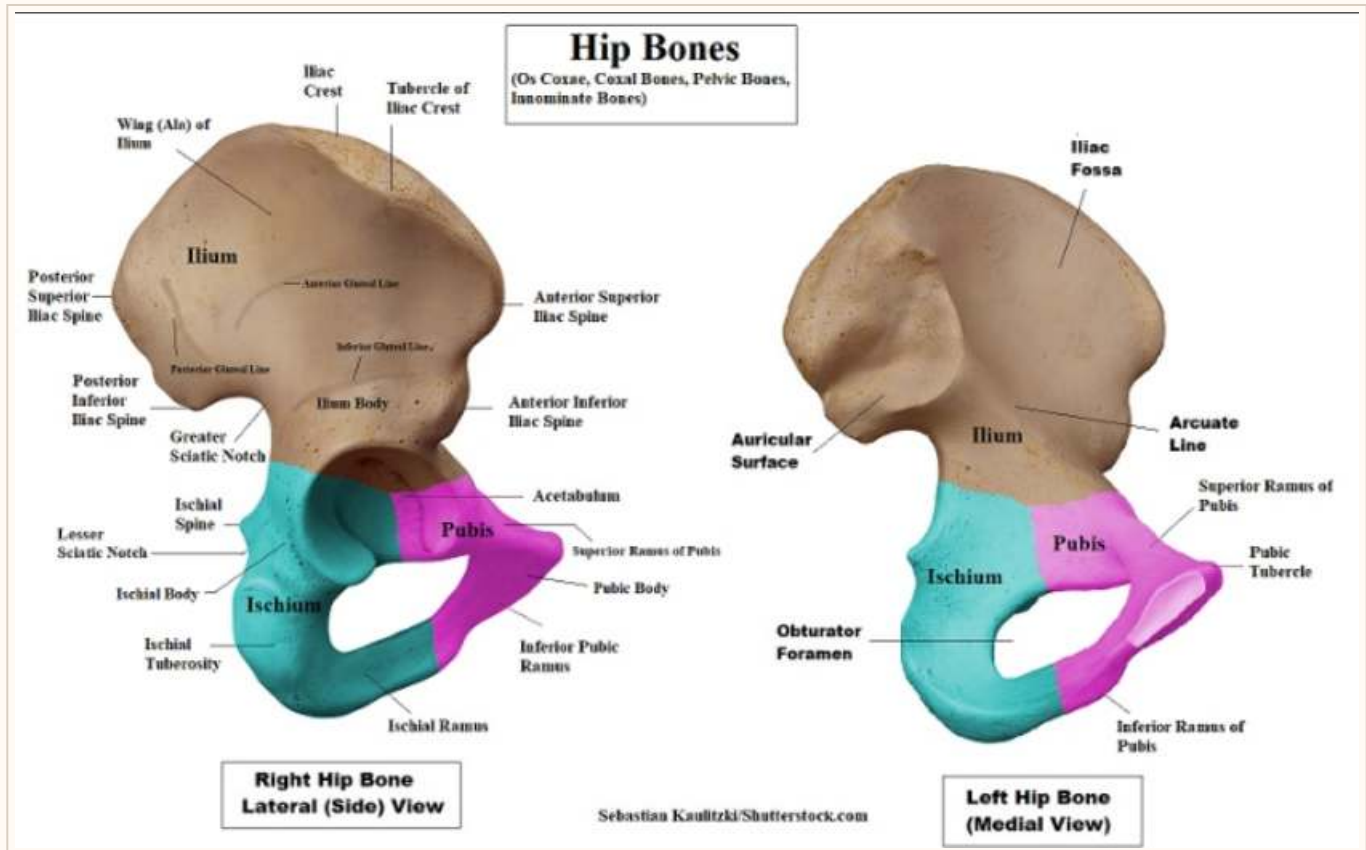
EXAM HACK — SCREW-HOME MECHANISM
The medial condyle is longer/more curved than the lateral condyle, causing the tibia to rotate laterally as the knee approaches full extension, "locking" the joint for standing with minimal muscular effort.

CLINICAL
Femoral neck fracture → shortened, externally rotated limb; retrograde blood supply via circumflex femoral arteries risks avascular necrosis. Supracondylar femoral fracture → risk to the popliteal artery (tethered at the adductor hiatus) and tibial/common peroneal nerves.

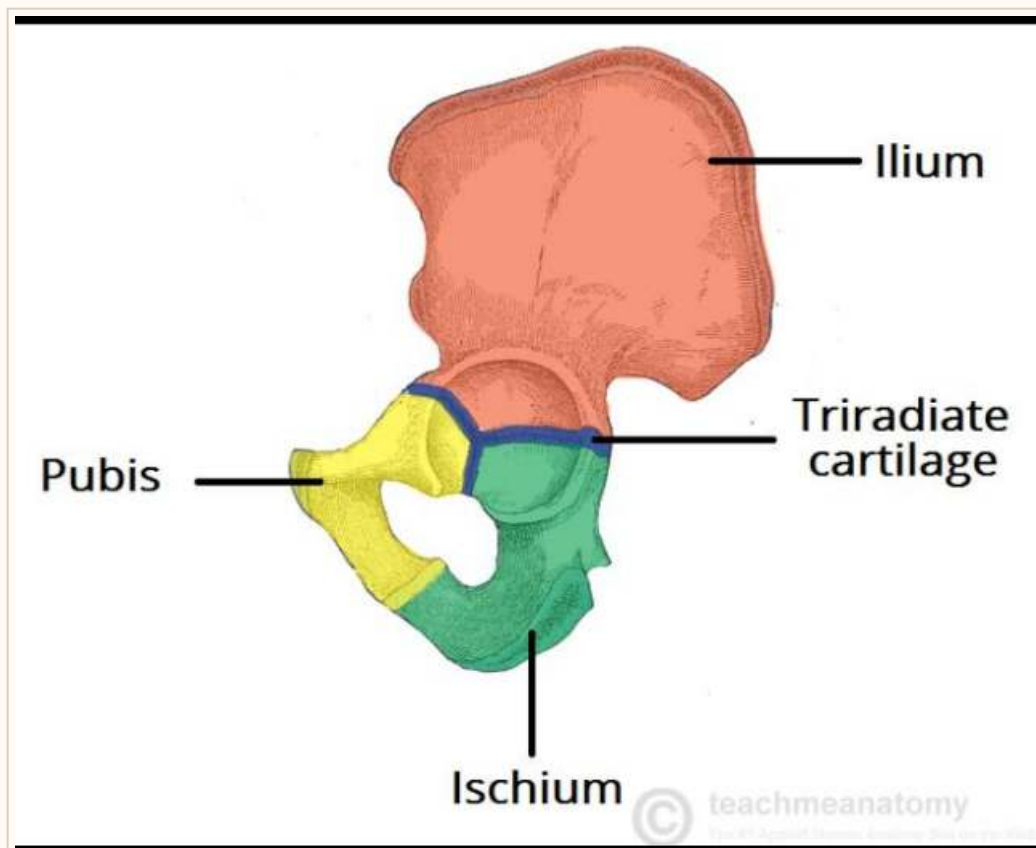
Patella, Tibia, and Fibula

OSTEOLOGY — HIP BONE (Reference Images)

Hip Bone landmarks — lateral & medial views



Ilium, Ischium, Pubis meeting at the triradiate cartilage



Pelvic Fracture — Duverney Fracture

Alzheimer's Mode: An isolated crack straight through the iliac wing — the "wing" bone breaks alone, while the weight-bearing pelvic ring stays intact.



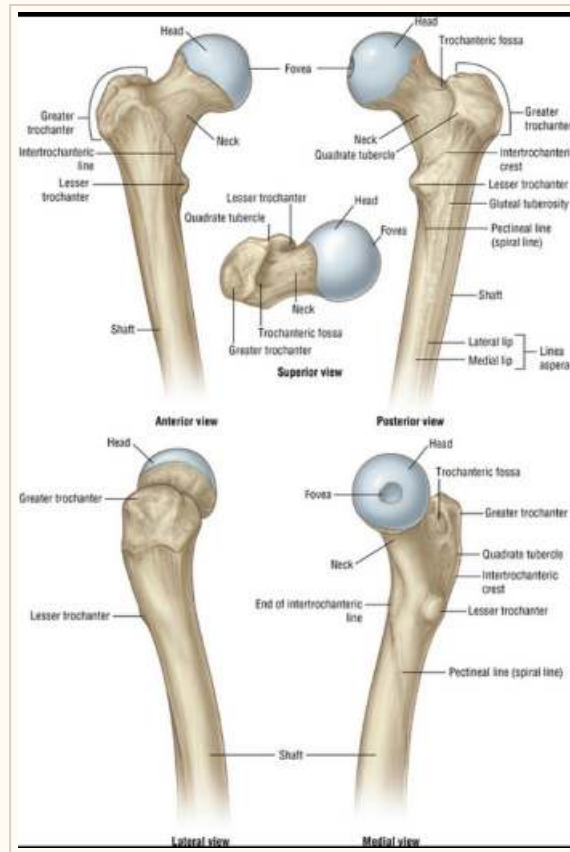
An isolated fracture of the **iliac wing/ala** (the fan-shaped superior part of the ilium), typically from a direct lateral blow to the hip. Classically considered a **stable pelvic fracture** because it does not disrupt the pelvic ring (no involvement of the acetabulum, obturator ring, or sacroiliac joint) — unlike ring-disrupting fractures (e.g. Malgaigne fracture) which are unstable.

EXAM HACK

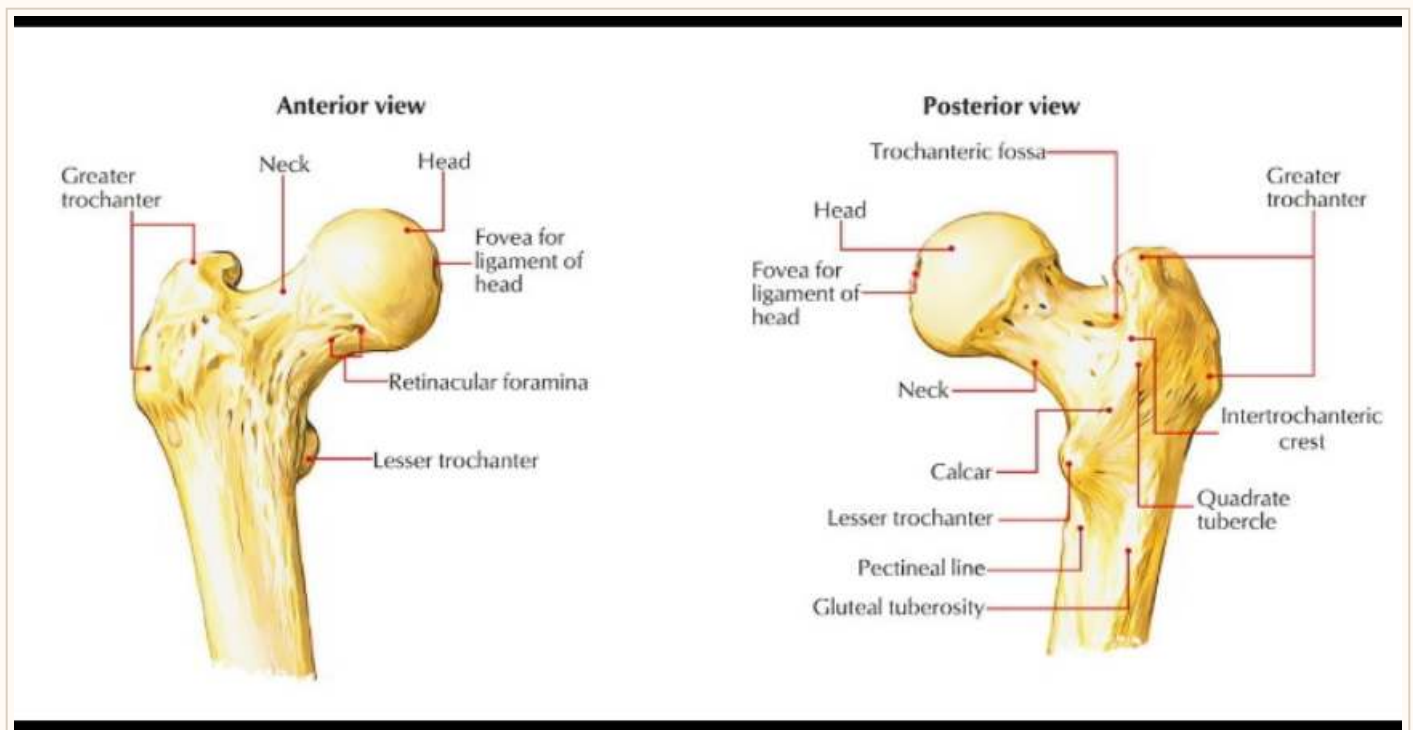
Duverney = iliac wing only, ring stays intact, generally managed conservatively. Contrast with pelvic ring fractures (e.g. from high-energy trauma), which are unstable and carry major hemorrhage risk from the presacral venous plexus.

OSTEOLOGY — FEMUR (Reference Images)

Proximal femur — head, trochanters, fovea, quadrate tubercle (3 views)



Femur anterior & posterior — calcar, retinacular foramina



Alzheimer's Mode: *The patella is the largest sesamoid bone, a "pebble" in the quadriceps tendon acting as a pulley. Tibia = weight-bearer, fibula = anchor point.*

Patella

A round-triangular bone, apex pointing inferiorly, the most anterior structure of the knee joint. Its ventral (anterior) surface is irregular/subcutaneous; its dorsal (posterior) surface is smooth and articular, meeting the femoral patellar surface. It is fixed superiorly within the quadriceps tendon (in which it is embedded) and continues inferiorly as the **patellar ligament**. The dorsal surface shows two facets — a smaller medial facet (meets the medial condyle) and a larger lateral facet (meets the lateral condyle). Function: prevents over-extension of the knee and increases the mechanical advantage (lever arm) of the quadriceps.

Tibia — Proximal Features

- **Tibial plateau** — the superior surface, showing the articular surfaces for the menisci.
- **Intercondylar eminence** — a raised bony area centrally in the intercondylar region, with **medial and lateral tubercles** arising at its sides that divide the intercondylar region into anterior and posterior parts. Attachment site for the cruciate ligaments and meniscal horns.
- **Condyles** — the broad medial and lateral proximal expansions giving the tibia its widened upper appearance.
- **Tibial tuberosity** — anterior, in the median plane, below the condyles; attachment for the patellar ligament.
- **Fibular facet** — posterolateral surface, just inferior to the lateral condyle, cartilage-lined, articulating with the fibular head (proximal tibiofibular joint).

Tibia — Shaft

- **Anterior border** — a sharp raised ridge running the length of the tibia, dividing the anterior part into medial and lateral surfaces (this is "the shin," subcutaneous and easily palpated).
- **Medial border** — a slightly elevated ridge separating the medial and posterior surfaces.
- **Interosseous border** — a sharp elevated ridge on the lateral surface, separating the lateral and posterior surfaces and anchoring the interosseous membrane.
- **Soleal line** — an oblique posterior ridge running inferomedially, attachment for several muscles (including part of soleus).

Tibia — Distal End

- **Medial malleolus** — a medial bony projection beyond the distal shaft, articulating (with the fibula) at the ankle joint.
- **Fibular notch** — a cartilage-covered lateral surface articulating with the distal fibula (distal tibiofibular joint).

Fibula

A thin bone lying lateral to the tibia, articulating with it proximally and distally.

- **Proximal articulating surface** — oblique, pointing inferomedially, forming the proximal tibiofibular joint.
- **Apex** — a bony projection above the articulating surface, attachment for ligaments/muscles.
- **Neck** — clinically significant: the **common (fibular) peroneal nerve** runs obliquely across its posterior surface here — the classic site of injury (foot drop).
- **Shaft** — three surfaces (medial, posterior, lateral); anterior border (sharp ridge) separates medial/lateral surfaces; interosseous border on the medial surface anchors the interosseous membrane.
- **Distal articulating surface** — forms the distal tibiofibular joint.
- **Lateral malleolus** — lateral bony projection at the ankle, with a **malleolar fossa** on its medial surface — attachment for the transverse tibiofibular and posterior talofibular ligaments.

The **interosseous membrane of the leg** extends from the interosseous border of the tibia to its counterpart on the fibula, separating the anterior and posterior leg muscle compartments.

EXAM HACK — FOREARM-LEG PARALLEL

The interosseous membrane's compartment-separating role in the leg directly mirrors its role in the forearm — same structural logic, different limb. Also: fibular neck = common peroneal nerve, exactly parallel to the humeral spiral groove = radial nerve theme from the upper limb.

Bones of the Foot

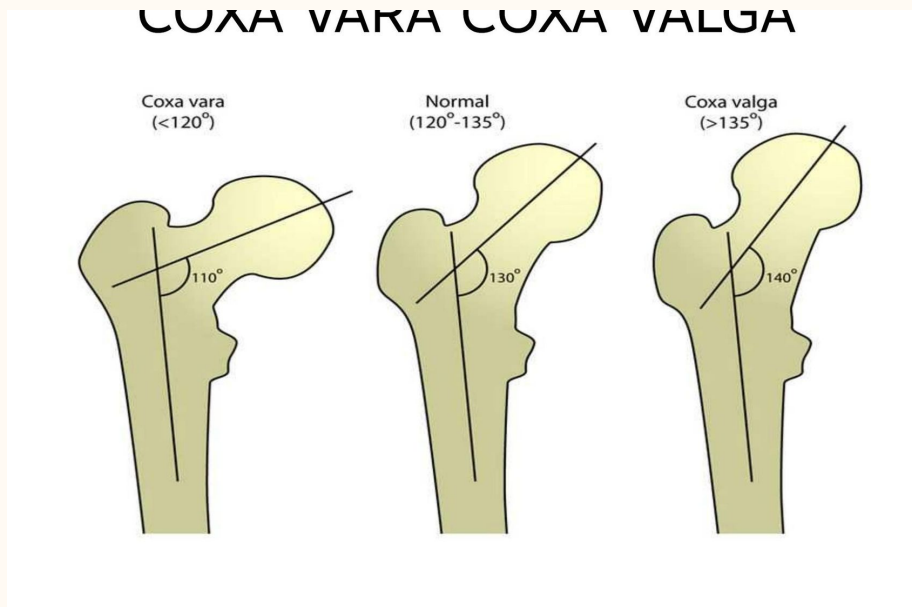
Alzheimer's Mode: *7 tarsals in 3 rows, forming a tripod with the calcaneus (heel) and metatarsal heads bearing weight.*

Tarsal Bones

- **Talus** — lies medially in the posterior (hindfoot) compartment; the only tarsal bone with no muscle attachments. A large raised cartilaginous body surface articulates with the tibia/fibula (ankle joint); a lateral articulating surface meets the calcaneus; a lateral process overlies part of the calcaneal body. Posteriorly it shows a medial and a lateral tubercle; anteriorly it articulates with the navicular. Its upper surface bears the **sustentaculum tali** — attachment for the tibiocalcaneal ligament.
- **Calcaneus** — lateral hindfoot bone, the largest tarsal, forms the heel. Articulates anteriorly with the cuboid and medially with the talus. On its medial side it also shows a sustentaculum tali, attachment for the plantar calcaneonavicular (spring) ligament.
- **Navicular** — medial midfoot, broad and rounded, anterior to the talus; articulates posteriorly with the talus, laterally with the cuboid, anteriorly with the three cuneiforms; has a tuberosity for the tibionavicular ligament.
- **Cuboid** — lateral midfoot, anterior to the calcaneus; articulates posteriorly with the calcaneus, medially with the navicular and lateral cuneiform, anteriorly with the 4th and 5th metatarsals.
- **Medial, Intermediate, Lateral cuneiforms** — the medial cuneiform is the largest, articulating with the 1st metatarsal; intermediate lies between the other two, articulating with the 2nd metatarsal; lateral cuneiform articulates with the 3rd metatarsal.

Femoral Neck-Shaft Angle — Coxa Vara & Coxa Valga

Alzheimer's Mode: Think of the neck-shaft angle as a hinge setting. Too closed (small angle) = coxa VARa ("vara" = varies inward/toward midline). Too open (large angle) = coxa VALga (valgus = away from midline).



Coxa Vara: $<120^\circ$ — neck-shaft angle reduced. Leg appears shortened, abductor lever arm shortened → Trendelenburg gait. Causes: fracture malunion, rickets, slipped capital femoral epiphysis (SCFE).

Normal: 120° – 135° (commonly cited as $\sim 125^\circ$).

Coxa Valga: $>135^\circ$ — neck-shaft angle increased. Associated with developmental dysplasia of the hip (DDH) and cerebral palsy; increases shear force on the femoral head.

Femoral Fracture Sites



Proximal femoral fractures are classified by site relative to the joint capsule:

Transcervical (neck) fracture — intracapsular; highest avascular necrosis risk since the retinacular vessels running up the neck are disrupted.

Intertrochanteric fracture — extracapsular, between greater and lesser trochanters; heavily vascularised bone — heals well, low AVN risk.

OSTEOLOGY — FEMUR (Complement)

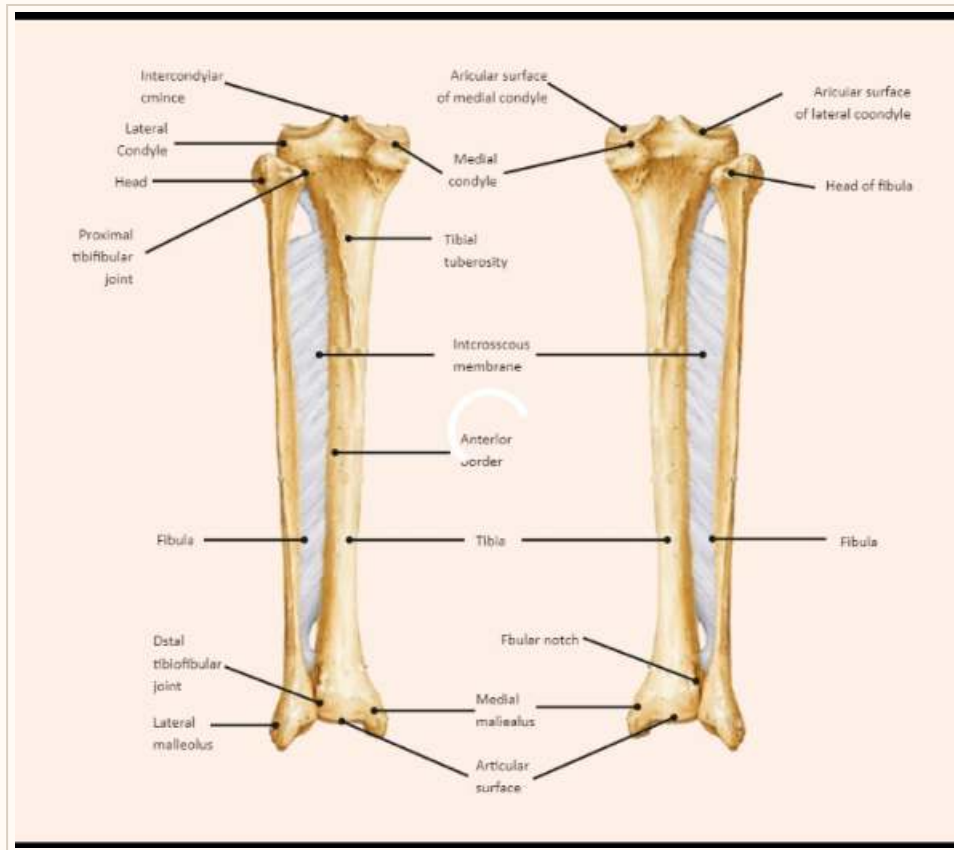
Subtrochanteric fracture — extracapsular, shaft just distal to the lesser trochanter; high mechanical stress zone, common in pathological fractures.

EXAM HACK

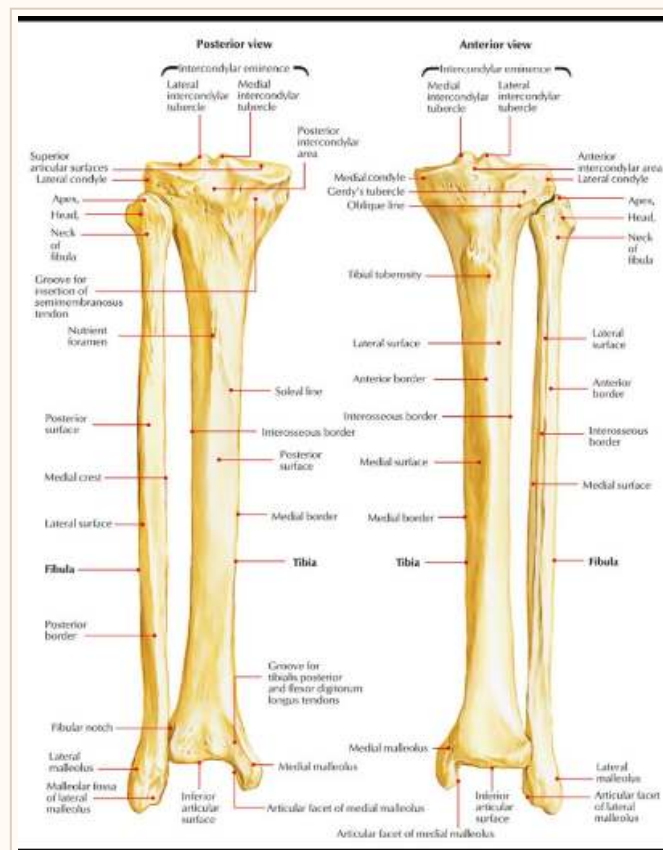
Intracapsular (transcervical/subcapital) fractures risk avascular necrosis because the retrograde blood supply via the medial/lateral circumflex femoral arteries is torn. Extracapsular fractures (intertrochanteric, subtrochanteric) largely spare this supply.

OSTEOLOGY — TIBIA & FIBULA (Reference Images)

Tibia & fibula bony landmarks



Tibia (shinbone) — anterior & posterior detail



Crus Fracture (Tibial/Fibular Shaft Fracture)

Alzheimer's Mode: "Crus" is just Latin for leg — a crus fracture is simply a break through the shaft of the tibia, fibula, or both, seen here on X-ray as the bright fracture line running obliquely through the bone.

CRUS FRACTURE



Fracture of the tibial and/or fibular shaft (the "crus" = leg segment between knee and ankle), usually from direct trauma, torsion (twisting falls, sports), or high-energy impact (road traffic accidents). The tibia is subcutaneous along its anteromedial border, so these fractures are often **open (compound)** fractures.

CLINICAL

Because the tibia lies within the tight anterior and deep posterior compartments, a crus fracture is a major risk factor for **acute compartment syndrome** — watch for the 6 P's (Pain out of proportion, Paraesthesia, Pallor, Pulselessness, Poikilothermia, Paralysis) and treat with emergency fasciotomy.

EXAM HACK

Mnemonic "Tiger Cubs Need MILC": Talus, Calcaneus, Navicular, Medial/Intermediate/Lateral cuneiforms, Cuboid. Talus has NO muscle attachments — all its movement is passive.

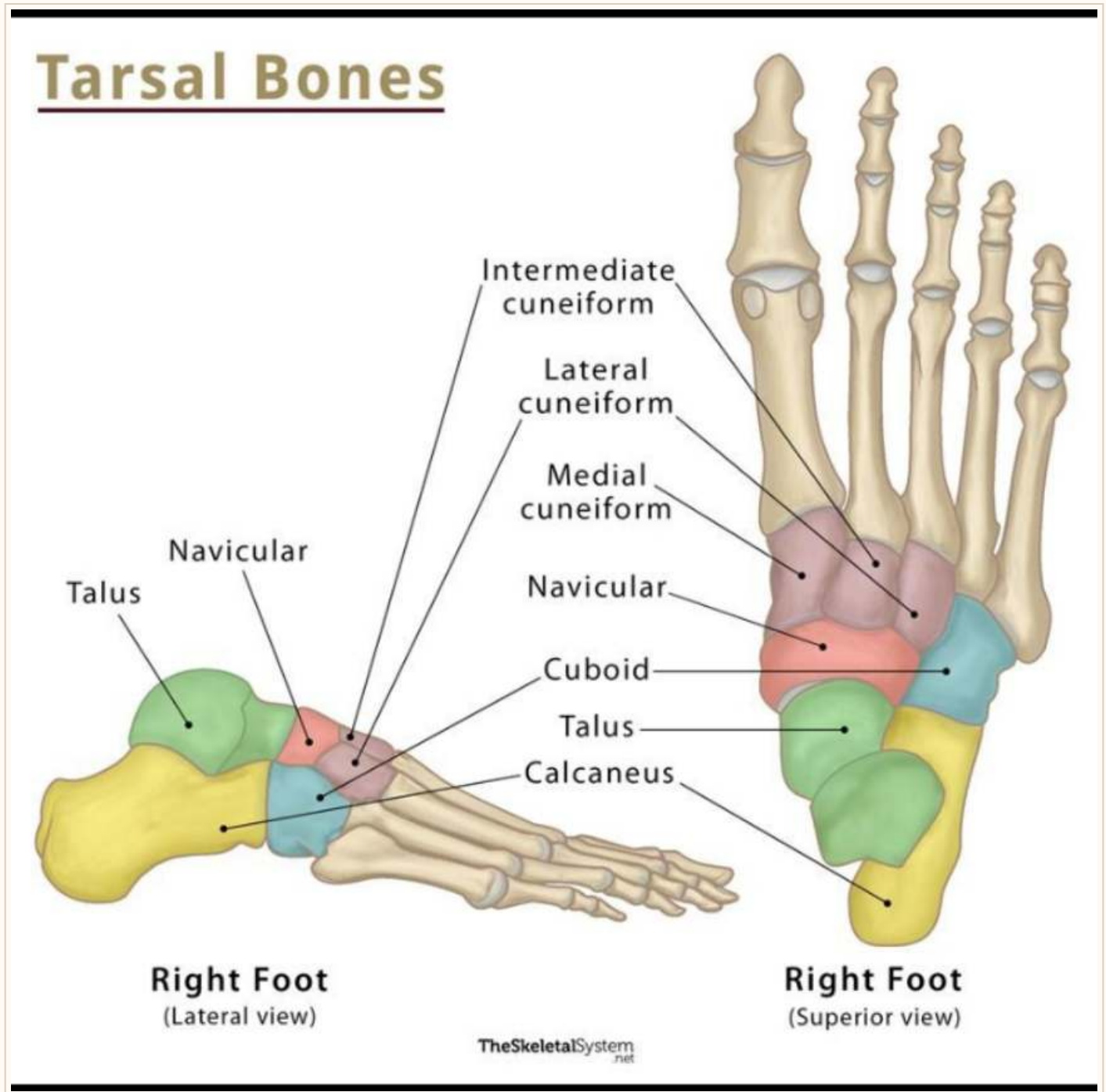
The **metatarsals** connect the tarsals to the phalanges, exactly like the metacarpals in the hand. Great toe: 2 phalanges. Toes 2-5: 3 phalanges each.

CLINICAL

Calcaneus fracture — fall from height, landing on the feet. Jones fracture — base of 5th metatarsal, poor healing zone. Talus fracture — retrograde blood supply risks avascular necrosis (same theme as scaphoid, femoral neck).

OSTEOLOGY — BONES OF THE FOOT (Reference Images)

Tarsal bones — lateral & superior views ("Tiger Cubs Need MILC")



2. JOINTS AND LIGAMENTS OF THE LOWER LIMB

Hip Joint

Alzheimer's Mode: Hip = ball-and-socket built for stability over mobility — the opposite trade-off from the shoulder.

Type: Synovial ball-and-socket, formed by the acetabulum and the femoral head.

- **Acetabular labrum** — fibrocartilaginous rim, deepens the socket.
- **Fibrous capsule** — covers the whole neck anteriorly, only half posteriorly (asymmetry relevant to fracture classification).
- **3 extracapsular ligaments** (spiral, tighten in extension): **Iliofemoral ("Y-ligament of Bigelow")** — strongest ligament in the body, prevents hyperextension; **Pubofemoral** — limits excessive abduction; **Ischiofemoral** — limits medial rotation/adduction, comparatively weakest → why posterior dislocation is most common.
- **Ligamentum teres (foveal ligament)** — intracapsular, carries the foveal artery to part of the femoral head.

EXAM HACK

All 3 extracapsular ligaments spiral so they tighten in extension — standing is maximum stability, minimum effort, the same "locking" logic as the knee's screw-home mechanism.

CLINICAL

Main blood supply to the femoral head (adults): medial + lateral circumflex femoral arteries (branches of profunda femoris), running retrograde up the neck — femoral neck fracture disrupts this → avascular necrosis. Posterior hip dislocation ("dashboard injury") is most common, given the ischiofemoral ligament/posterior capsule weakness. Trendelenburg sign: hip abductor weakness → pelvis drops on the OPPOSITE side during single-leg stance.

Knee Joint

Alzheimer's Mode: Knee = hinge joint with almost no bony stability (flat tibial plateau meets rounded femoral condyles) — held together entirely by ligaments, menisci, and bursae.

Type: Synovial hinge (modified — allows slight rotation). Composed of: distal femur, medial + lateral menisci, proximal tibia, the patella, multiple bursae, and numerous ligaments.

Menisci

Semi-circular, C-shaped fibrocartilaginous discs sitting atop their respective tibial articulatory surfaces, providing a frictionless gliding surface for the femoral condyles during flexion.

Bursae of the Knee (4 total)

Bursa	Location
Semimembranous bursa	Posterior compartment of the knee
Infrapatellar bursa (x2)	Inferior to the patella — one lies subcutaneously anterior to the patellar ligament, the other lies posterior (deep) to it
Prepatellar bursa	Overlies the patella, subcutaneous
Suprapatellar bursa	Superior to the patella, posterior (deep) to the quadriceps femoris tendon

ADDED DETAIL — KNEE BURSAE

This is a full 4-bursa system distinct from (and in addition to) the popliteal (Baker's) cyst concept — the semimembranous, infrapatellar (x2), prepatellar, and suprapatellar bursae each reduce friction at a different point around the joint. Prepatellar bursitis ("housemaid's knee") and infrapatellar bursitis ("clergyman's knee") are named clinical entities tied directly to these two subcutaneous bursae.

Ligaments of the Knee

- **Transverse ligament** — connects the medial and lateral menisci anteriorly.
- **Medial collateral ligament (MCL)** — superior surface of the medial epicondyle → medial tibial condyle; gives off a band blending with the knee capsule; firmly attached to the medial meniscus.
- **Lateral collateral ligament (LCL)** — anterolateral surface of the lateral epicondyle → the fibular head's designated depression; NOT attached to the lateral meniscus.
- **Oblique popliteal ligament** — flat, broad, from the lateral epicondyle, curving posteriorly around the knee to the medial tibial condyle; contributes to the floor of the popliteal fossa and has small openings through which nerves/vessels pass.
- **Anterior cruciate ligament (ACL)** — posterior surface of the medial intercondylar wall (in the intercondylar fossa) → anterior proximal tibia.
- **Posterior cruciate ligament (PCL)** — anterior surface of the lateral intercondylar wall → posterior proximal tibia.

Beneath these ligaments, the knee is enclosed by a fibrocartilaginous **knee joint capsule**, with the **synovial membrane** lying beneath that.

ADDED DETAIL — OBLIQUE POPLITEAL LIGAMENT

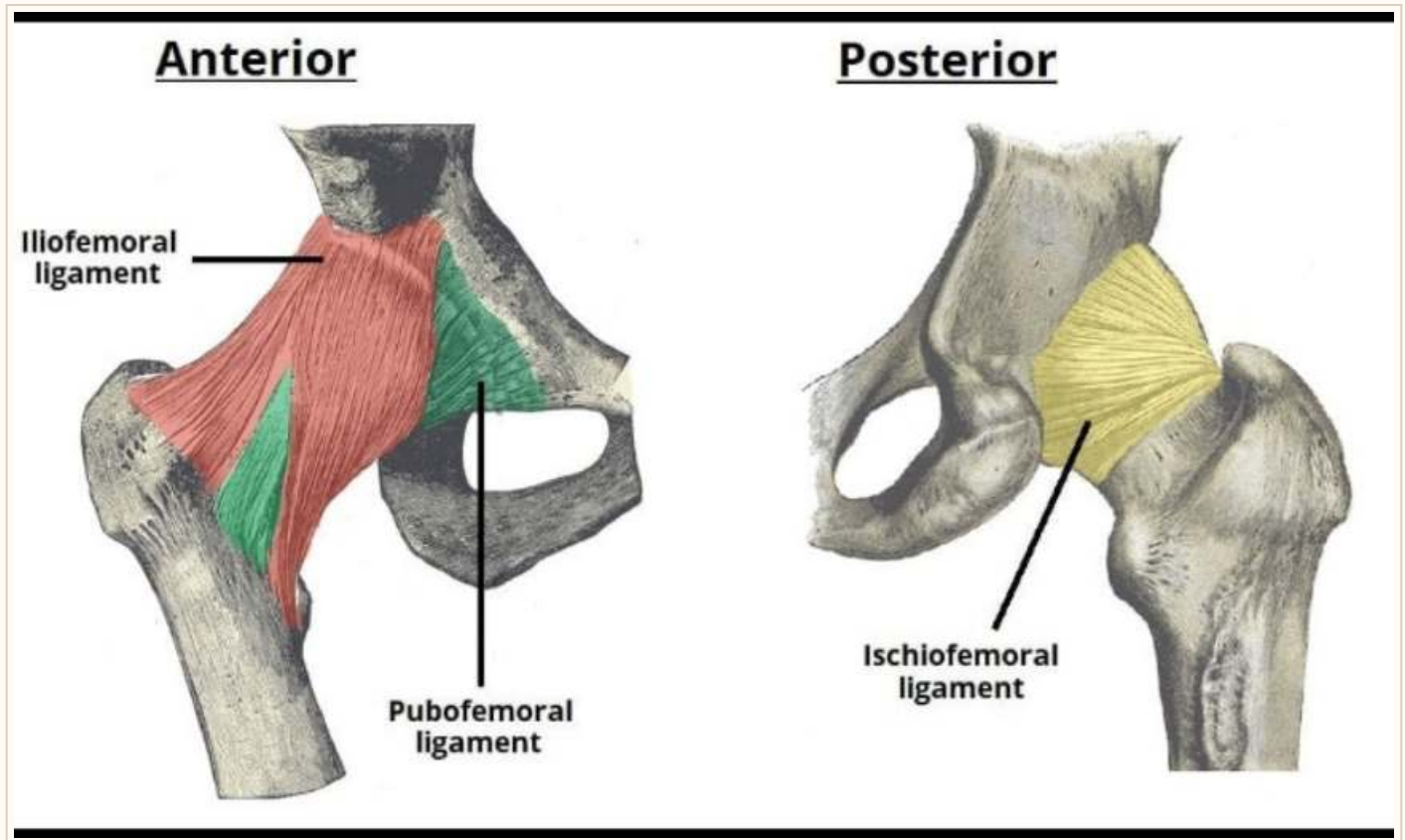
This ligament, arising from the lateral epicondyle and sweeping posteromedially to the tibia, is a distinct reinforcement of the posterior knee capsule that directly contributes to forming the floor of the popliteal fossa — worth naming explicitly alongside the popliteus/gastrocnemius muscular boundaries already covered.

EXAM HACK — RECAP

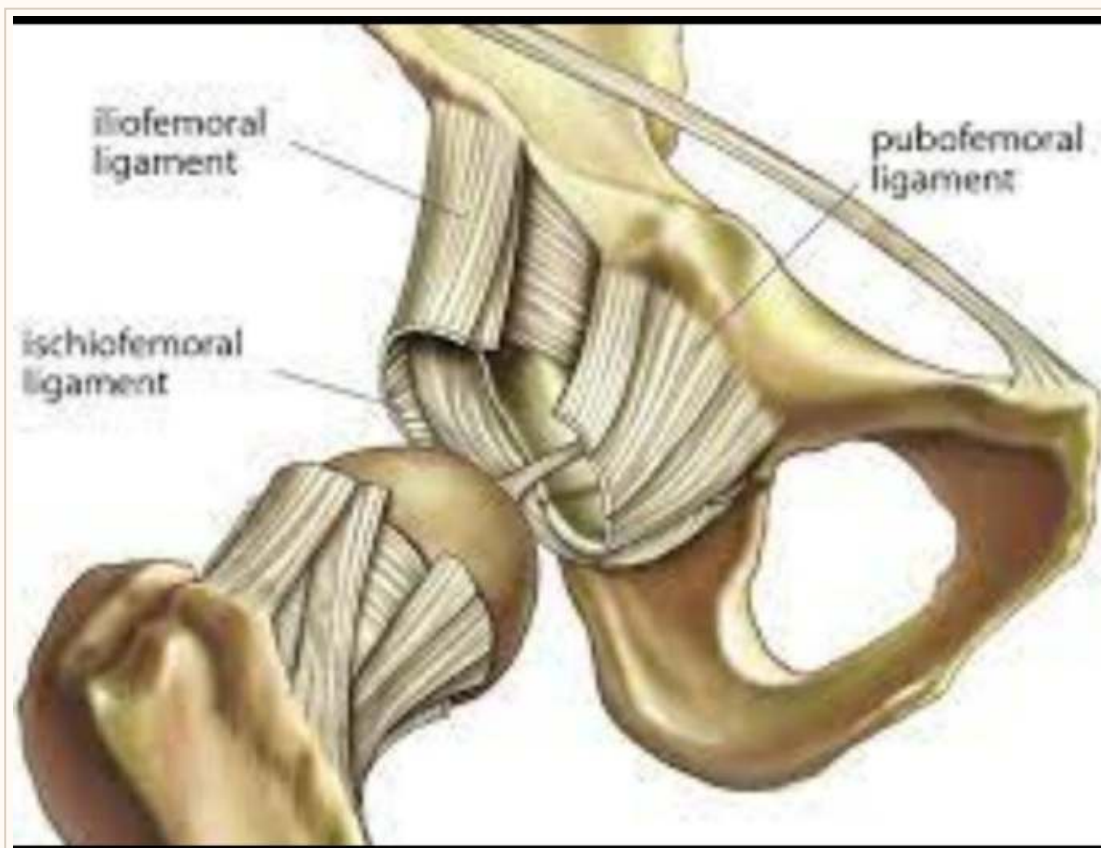
MCL attached to medial meniscus (injured together); LCL NOT attached to lateral meniscus. ACL/PCL named by TIBIAL attachment. "Terrible triad" = MCL +

JOINTS & LIGAMENTS — HIP JOINT (Reference Images)

Iliofemoral, pubofemoral & ischiofemoral ligaments — anterior/posterior

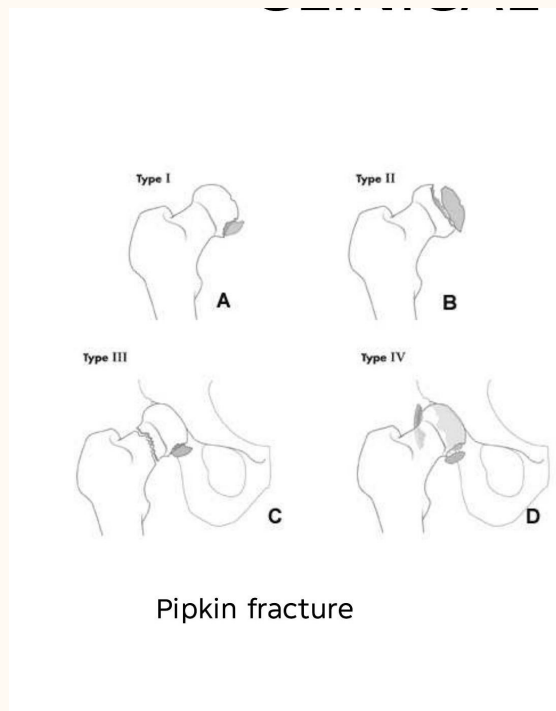


Hip joint ligaments — 3D view



Pipkin Fracture (Femoral Head Fracture)

Alzheimer's Mode: Pipkin is what happens when a posterior hip dislocation "shaves off" a piece of the femoral head against the acetabular rim on the way out — the 4 types just describe how much of the head breaks and whether the neck/acetabulum go with it.



A fracture of the femoral head occurring in association with a **posterior hip dislocation** (the classic "dashboard injury" already covered under Femur). Classified into 4 types:

Type I — fracture inferior to the fovea capitis (below the ligamentum teres attachment).

Type II — fracture superior to the fovea capitis (larger fragment, weight-bearing surface involved).

Type III — Type I or II plus an associated femoral neck fracture.

Type IV — Type I or II plus an associated acetabular rim fracture.

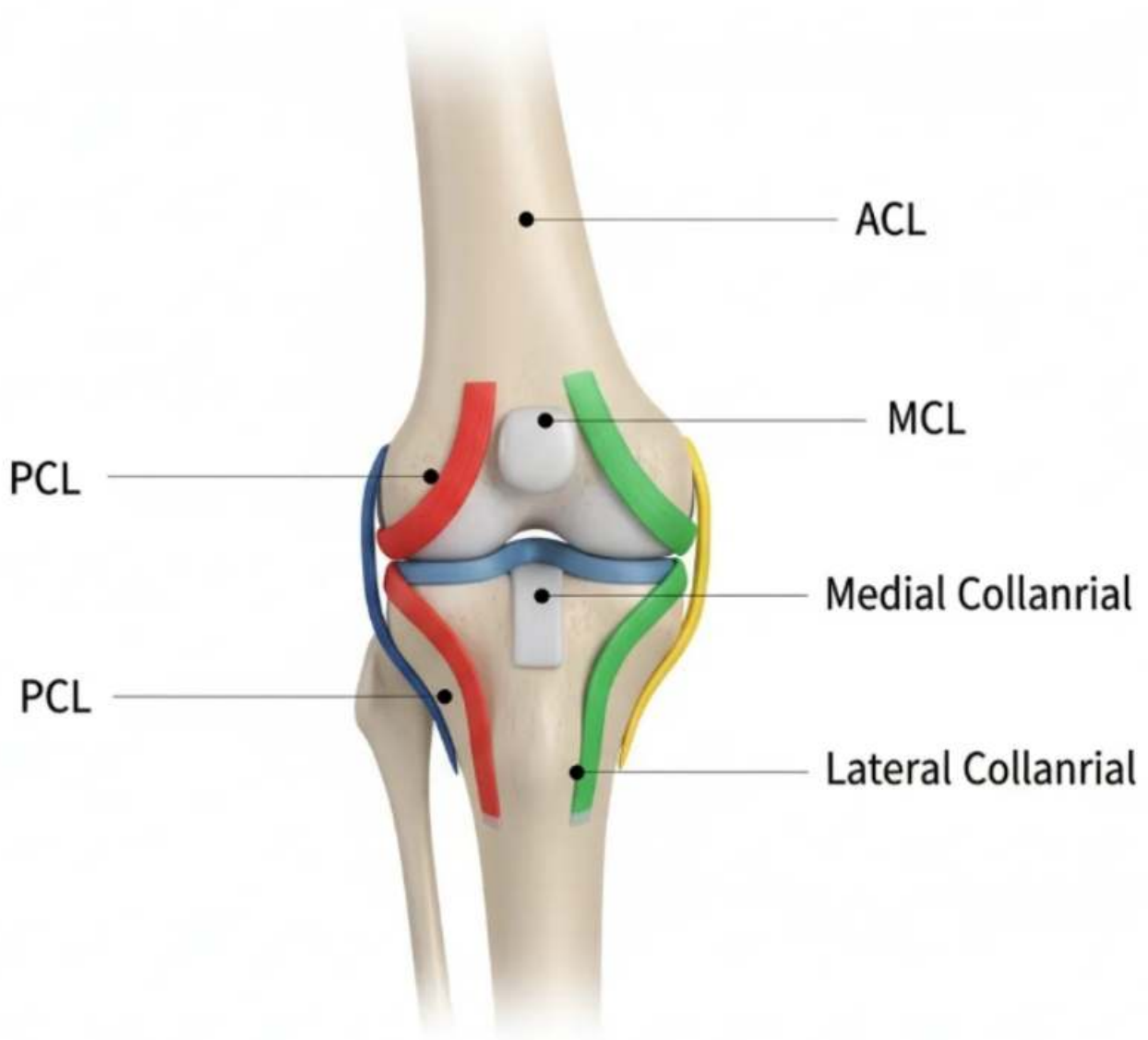
EXAM HACK

Pipkin fracture = femoral HEAD fracture + posterior dislocation. Don't confuse with a plain femoral neck fracture (no dislocation) or with Duverney (isolated iliac wing, no hip joint involvement at all).

JOINTS & LIGAMENTS — KNEE JOINT (Reference Images)

Four main knee ligaments — ACL, PCL, MCL, LCL

Four Main Knee Ligaments: ACL, PCL, MCL, LCL



ACL + medial meniscus from a lateral blow to a planted, slightly flexed knee. Dashboard injury (posterior force on flexed knee) → classic PCL tear.

Ankle Joint

Alzheimer's Mode: *A bony mortise (tibia + fibula) gripping a wedge-shaped talus, snuggest in dorsiflexion (most stable), loosest in plantarflexion (easiest to sprain).*

Type: Synovial hinge. Bony mortise: medial malleolus (tibia) + lateral malleolus (fibula, longer) + posterior malleolus (tibia), gripping the talus.

Lateral Ligaments (3 separate bands)

- **Anterior talofibular ligament (ATFL)** — anterior margin of lateral malleolus → upper surface of talus; stabilizes against anterior dislocation. Weakest, **most commonly injured ligament in the entire body**.
- **Posterior talofibular ligament (PTFL)** — malleolar fossa of fibula → medial tubercle on the posterior talus; stabilizes against posterior dislocation. Strongest, rarely injured.
- **Calcaneofibular ligament (CFL)** — posteroinferior margin of lateral malleolus → lateral tubercle of the calcaneus; stabilizes against inward (inversion) dislocation.

Medial Ligament Complex ("Deltoid Ligament") — 4 named bands

- **Tibionavicular ligament** — anteroinferior margin of medial malleolus → tuberosity of the navicular; stabilizes, prevents anterior/outward (eversion) ankle dislocation.
- **Tibiocalcaneal ligament** — inferior margin of medial malleolus → sustentaculum tali of the calcaneus; opposes/complements the tibionavicular ligament's action.
- **Anterior tibiotalar ligament** — anterior margin of medial malleolus → upper surface of the distal talus; lies medial to the tibionavicular ligament, opposing its function.
- **Posterior tibiotalar ligament** — posteroinferior margin of medial malleolus → medial tubercle of the talus; stabilizes against posterior dislocation.

ADDED DETAIL — THE DELTOID LIGAMENT HAS 4 NAMED PARTS

Earlier teaching treated "the deltoid ligament" as one broad band. It is actually a fan-shaped complex of **4 distinct named ligaments** — tibionavicular, tibiocalcaneal, anterior tibiotalar, and posterior tibiotalar — each with its own precise origin/insertion and stabilizing role. This full breakdown is exactly the kind of detail that separates a pass from a "very dangerous" exam performance.

EXAM HACK

Inversion sprain → ATFL injured first, then CFL if force continues, PTFL almost never. Eversion injury → deltoid ligament strain OR (more often) an avulsion fracture, since the deltoid complex is stronger than any single lateral band.

Ligaments of the Sole of the Foot

Alzheimer's Mode: *The sole has its own dedicated ligament system supporting the plantar arches — distinct from the ankle ligaments above.*

- **Short plantar ligament** — calcaneal tuberosity (distal sole surface) → sole surface of the cuboid.
- **Long plantar ligament** — proximal sole surface of the calcaneus → runs anteriorly, dividing at the cuboid into 4 bands terminating on the bases of the first four metatarsals.
- **Plantar calcaneonavicular ligament ("spring ligament")** — medial/distal sole surface of the calcaneus (from the sustentaculum tali) → sole surface of the navicular; directly supports the head of the talus and the medial longitudinal arch.

ADDED DETAIL — SOLE LIGAMENTS WERE MISSING ENTIRELY

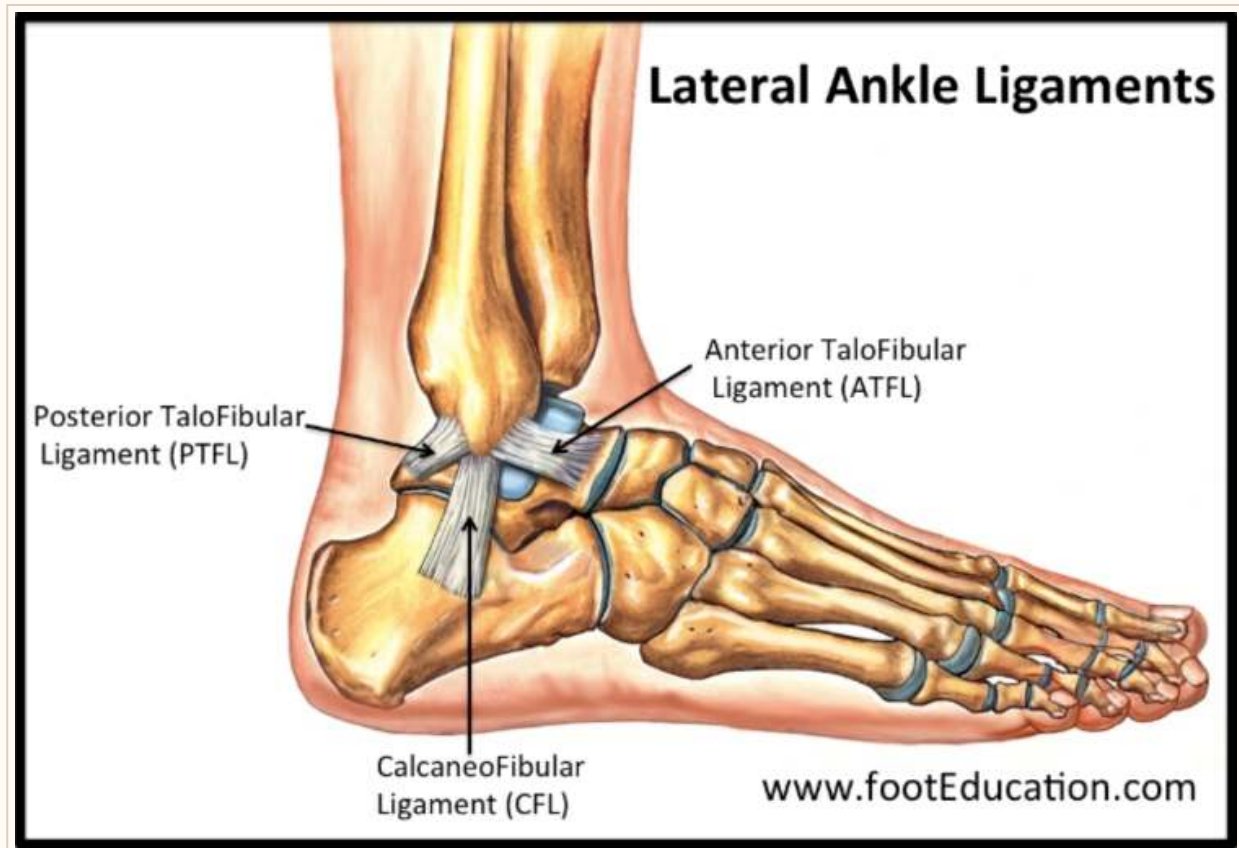
These three plantar ligaments (short plantar, long plantar, and the plantar calcaneonavicular/"spring" ligament) form the ligamentous support system of the foot's arches and were not covered in the earlier teaching pass at all. The spring ligament in particular is a classic exam structure — its laxity/failure is a major contributor to acquired flatfoot deformity, working alongside tibialis posterior (already covered in musculature).

CLINICAL

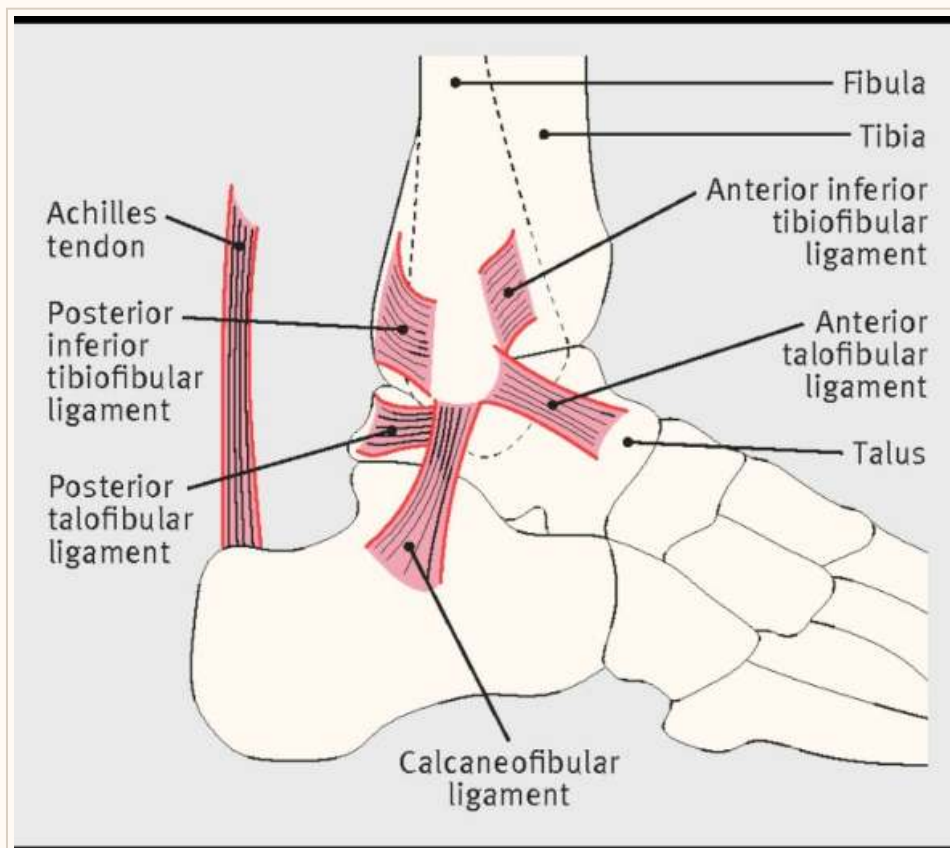
High ankle sprain (syndesmotic injury) — different from typical lateral sprains, tested with the squeeze test, longer recovery. Ottawa ankle rules — clinical decision tool for imaging need. Spring ligament failure + tibialis posterior dysfunction together are the two major structural causes of adult-acquired flatfoot.

JOINTS & LIGAMENTS — ANKLE JOINT (Reference Images)

Lateral ankle ligaments — ATFL, PTFL, CFL

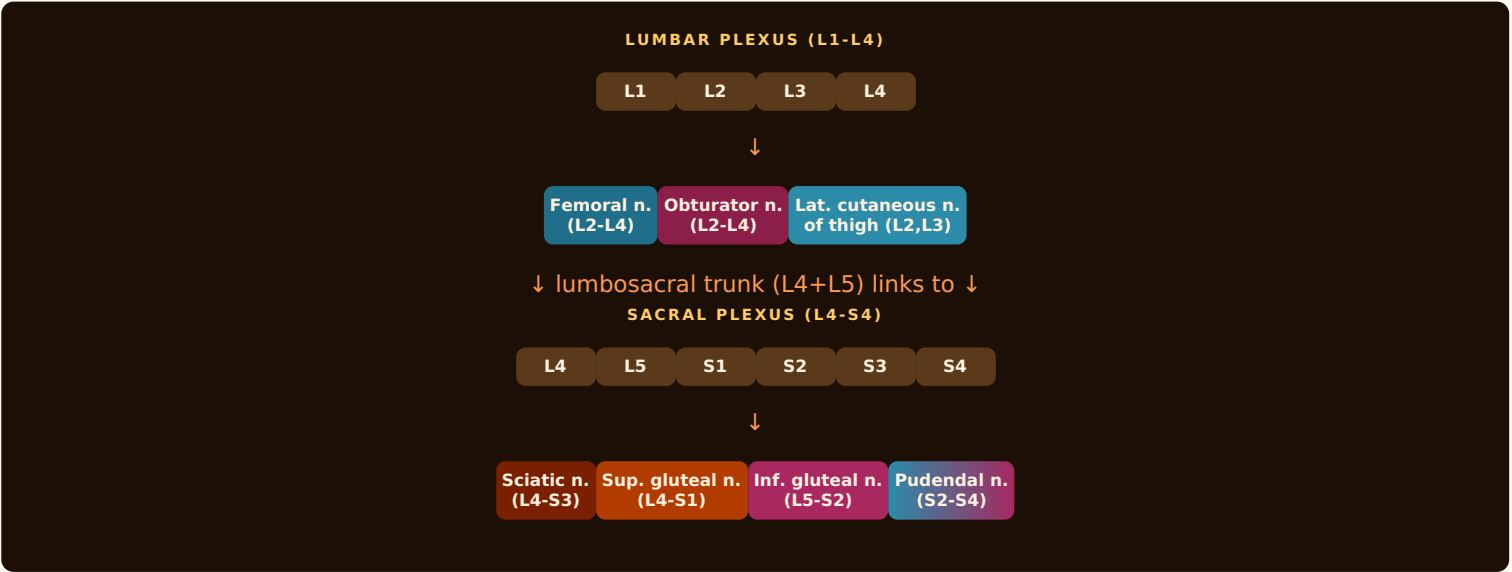


Ankle ligaments incl. tibiofibular syndesmosis



3. NERVE SUPPLY — THE LUMBOSACRAL PLEXUS

Alzheimer's Mode: Unlike the brachial plexus's single continuous pipeline, the lumbosacral plexus is really **two separate plexuses sharing one connecting cable** — the **lumbar plexus** (L1-L4, sits within psoas major) and the **sacral plexus** (L4-S4, sits on the posterior pelvic wall), linked by the **lumbosacral trunk**.



Motor Nerves

Nerve	Root	Motor territory
Femoral nerve	L2-L4	Anterior thigh (quadriceps, sartorius, pectineus); enters lateral to the femoral artery, deep to the inguinal ligament
Obturator nerve	L2-L4	Medial thigh (adductors); descends through the obturator canal
Sciatic nerve	L4-S3	Posterior thigh (hamstrings) then splits into tibial + common peroneal, supplying the whole leg/foot below the knee
Superior gluteal nerve	L4-S1	Gluteus medius, minimus, tensor fasciae latae
Inferior gluteal nerve	L5-S2	Gluteus maximus
Common peroneal nerve	(from sciatic)	Splits into deep peroneal (anterior leg) + superficial peroneal (lateral leg) around the fibular neck
Tibial nerve	(from sciatic)	Posterior leg; enters the foot posterior to the medial malleolus, splitting into medial + lateral plantar nerves

EXAM HACK

Femoral nerve = front of thigh (quads, matches anterior thigh musculature). Obturator nerve = inner thigh (adductors). Sciatic nerve is really the tibial + common peroneal nerves bundled together from the pelvis to the popliteal fossa, where they typically split — a high sciatic injury (hip level) knocks out BOTH; a lower injury affects only one branch's territory.

Full Cutaneous Nerve Map

Alzheimer's Mode: Every patch of skin on the lower limb has a named sensory nerve — this map is exactly what's tested when a question describes numbness in a specific region and asks which nerve is compressed.

Cutaneous Supply of the Thigh

Nerve	Root	Territory
Subcostal nerve	T12	Lateral surface of the superior thigh, just beneath the inguinal ligament
Femoral branch of the genitofemoral nerve	L1, L2	Medial surface of the superior thigh below the inguinal ligament, medial to the subcostal territory
Ilioinguinal nerve	L1	Anteromedial thigh, adjacent to the genitals
Lateral cutaneous nerve of thigh	L2, L3	Anterolateral + posterolateral thigh
Anterior cutaneous nerve of thigh (from femoral n.)	L2, L3	Anterior thigh, medial to the lateral cutaneous nerve's territory, plus anterolateral knee
Obturator nerve (cutaneous branch)	L2-L4	Medial + posteromedial thigh
Posterior cutaneous nerve of thigh	Posterior branches of S1-S3	Lower buttock/gluteal fold, posterior thigh, posterolateral + posterior knee
Posterior branches of S1-S3	S1-S3	Medial and inner buttock skin
Posterior branches of L1-L3	L1-L3	Remainder of the buttock skin (outside the S1-S3 and lower gluteal fold territories)

Cutaneous Supply of the Leg

Nerve	Root/Origin	Territory
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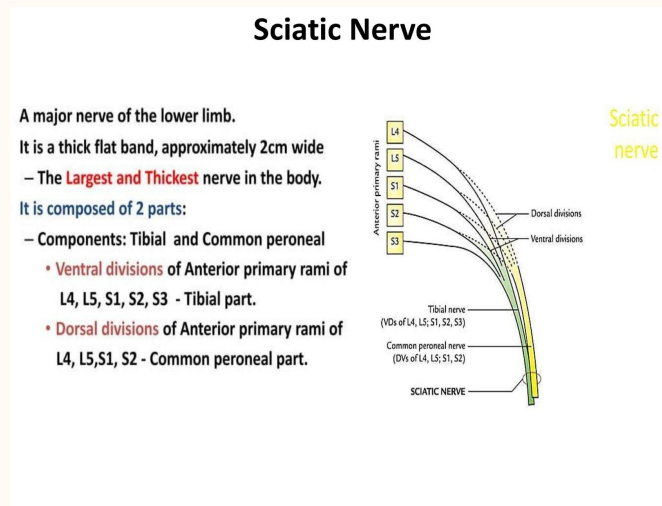
Sciatic Nerve — Dual Composition & Course

Alzheimer's Mode: The sciatic nerve is really two nerves (tibial + common peroneal) bundled together in one sheath from birth — they only "split for real" at the popliteal fossa, but you can always peel them apart because they come from different halves of the same spinal roots.

The **largest and thickest nerve in the body** (~2cm wide), formed from the sacral plexus (L4–S3) and composed of two functionally distinct parts bundled in a common sheath:

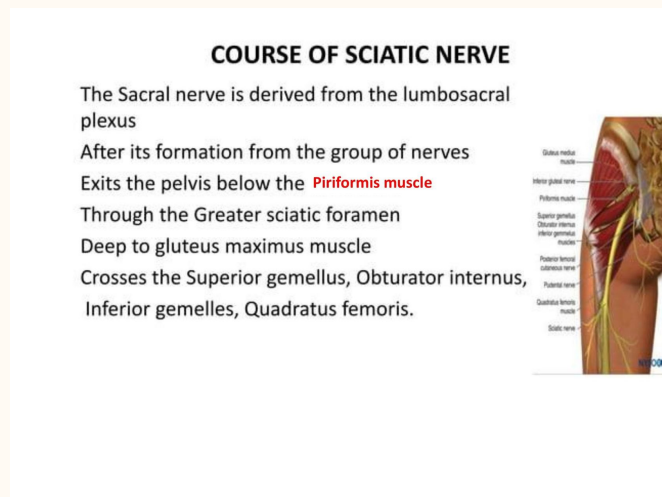
Tibial part — from the **ventral divisions** of the anterior primary rami of L4, L5, S1, S2, S3.

Common peroneal part — from the **dorsal divisions** of the anterior primary rami of L4, L5, S1, S2.



Course

Exits the pelvis through the **greater sciatic foramen**, below (or occasionally through/above) the **piriformis muscle**, then runs deep to gluteus maximus, crossing successively the superior gemellus, obturator internus, inferior gemellus, and quadratus femoris before descending into the posterior thigh.

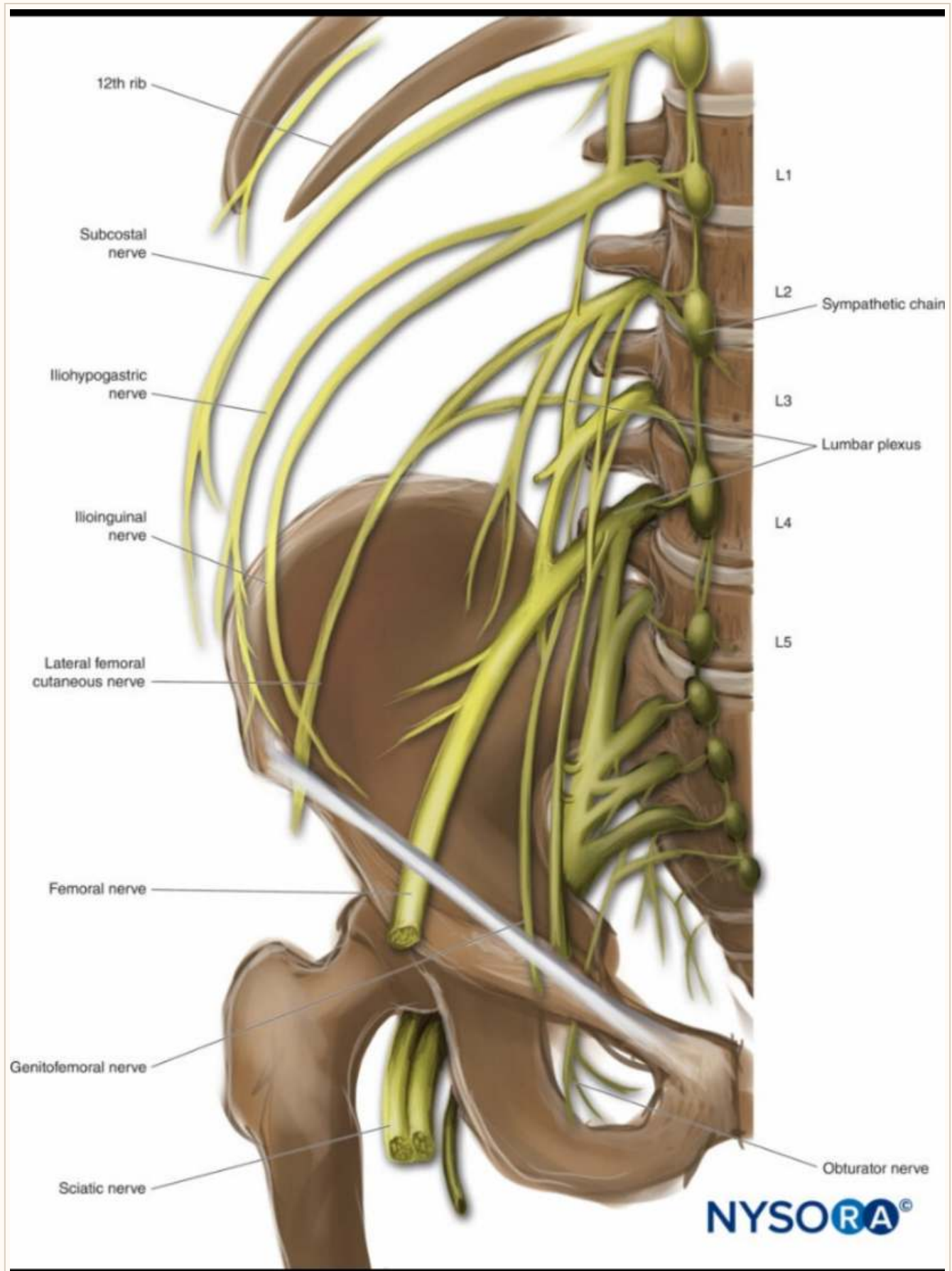


EXAM HACK

The sciatic nerve gives **no direct cutaneous branches** — all its sensory territory (posterolateral/anterolateral leg, sole of foot via the tibial nerve; lateral leg + dorsum of foot via the common peroneal nerve) is supplied only through its terminal branches. A useful trap: don't credit the sciatic nerve itself with any skin patch on a dermatome map.

NERVE SUPPLY — LUMBAR PLEXUS (Reference Image)

Lumbar plexus — subcostal, ilioinguinal, genitofemoral, femoral, obturator, sciatic nn.



Lateral cutaneous nerve of leg	From common peroneal n. (L5-S3)	Anterolateral, lateral, posterolateral leg
Saphenous nerve	From femoral n. (L3, L4)	Anterior, medial, posteromedial leg down to the medial malleolus; also medial knee
Posterior cutaneous nerve of thigh (extension)	—	Upper medial posterior leg
Superficial branch of common peroneal nerve	L4-S1	Anterolateral leg
Sural nerve	From tibial n. (L5-S2)	Lower posterior, posterolateral, and lateral leg (down to the lateral malleolus)

Cutaneous Supply of the Foot

Nerve	Territory
Superficial branch of common peroneal nerve	Dorsum of the foot, EXCEPT the web space between the 1st and 2nd toes
Deep branch of common peroneal nerve	Web space between the 1st and 2nd toes only
Sural nerve	Lateral side of the foot + lateral half of the 5th toe
Tibial nerve	Posterior heel/plantar surface generally
Medial calcaneal nerve (from sural n.)	Skin below the calcaneal (Achilles) tendon + posterior heel
Lateral plantar nerve	Lateral sole + tip of the 5th toe + lateral half of the 4th toe
Medial plantar nerve	Medial sole + medial 3½ toes (1st, 2nd, 3rd, and medial half of 4th) and their tips

ADDED DETAIL — FULL CUTANEOUS MAP WAS MISSING
The earlier teaching pass covered motor nerve territories thoroughly but only briefly touched cutaneous distribution. This complete thigh/leg/foot sensory map — including the subcostal nerve, genitofemoral branch, ilioinguinal nerve, and the precise dorsal/plantar foot boundaries — is exactly the level of detail needed to answer "numbness in region X, which nerve?" style questions with full confidence.

CLINICAL — ERB'S/KLUMPKE'S-EQUIVALENT CORRELATIONS
Meralgia paresthetica — compression of the lateral cutaneous nerve of thigh under the inguinal ligament near the ASIS → burning/tingling anterolateral thigh (obesity, pregnancy, tight clothing). **Long thoracic-equivalent here is Trendelenburg** — superior gluteal nerve injury → gait abnormality. **Sciatic nerve injury** (posterior hip dislocation, deep IM injection error) → combined hamstring + everything-below-knee deficit. **Piriformis syndrome** — sciatic nerve compressed by piriformis (or its anatomical variant), mimicking sciatica without true root pathology.

4. MUSCULATURE OF THE LOWER LIMB

Lower limb muscles are grouped by region: gluteal (superficial + deep), thigh (medial, anterior, posterior compartments), leg (anterior, lateral, posterior compartments), and the intrinsic foot muscles (dorsal + 4 plantar layers).

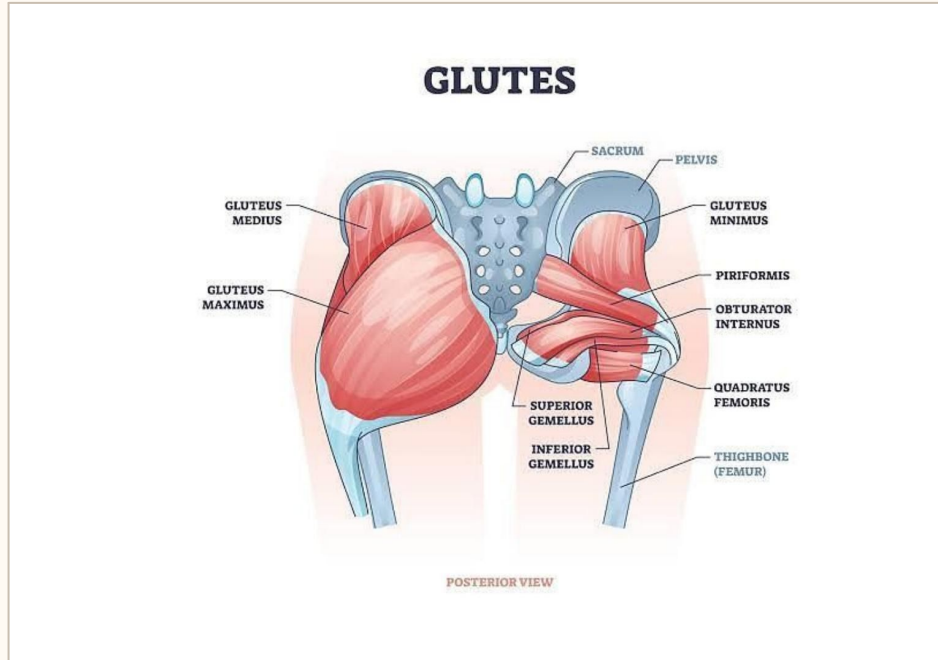
Gluteal Region

Alzheimer's Mode: Superficial layer = 3 big glutes (power + stabilization); deep layer = 6 small lateral rotators (fine control), mirroring the rotator cuff concept at the hip.

Superficial Group

Muscle	Origin	Insertion	Action	Nerve
Gluteus maximus	Lateral sacrum/coccyx, posterosuperior iliac crest, sacrotuberous ligament	Gluteal tuberosity of femur + posterior margin of IT tract	Stabilizes the pelvis; extension, abduction, lateral rotation of the thigh	Inferior gluteal nerve
Gluteus medius	Lateral 2/3 of the dorsal posterosuperior iliac crest (between the posterior and anterior gluteal lines), below gluteus maximus's origin	Superior + lateral surfaces of the greater trochanter	Stabilizes the hip, prevents contralateral hip drop; abduction + internal rotation	Superior gluteal nerve
Gluteus minimus	Dorsal ilium below gluteus medius's origin (between anterior and inferior gluteal lines)	Anterior surface of the greater trochanter	Same stabilizing role as gluteus medius; abduction + internal rotation	Superior gluteal nerve
Tensor fasciae latae	Lateral ASIS + lateral iliac crest tubercle	Anterior margin of the IT tract	Stabilizes the knee in extension; flexion + internal rotation of the femur, external rotation of the leg (via IT tract)	Superior gluteal nerve

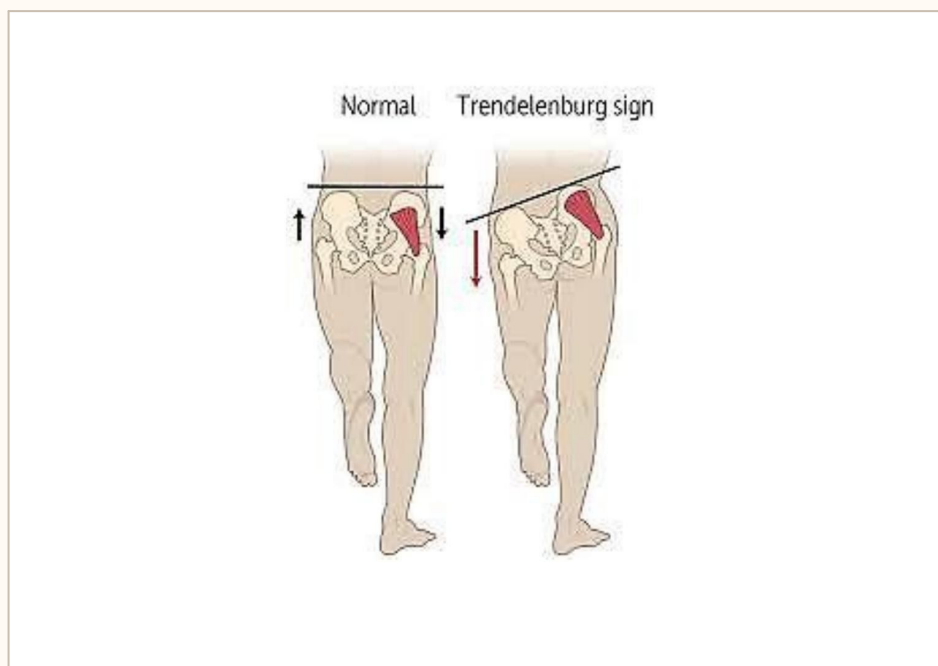
All gluteal muscles — composite labeled view (superficial + deep)



Reference image — supplementary visual aid, INDIGENE Study Console

Clinical correlate: Trendelenburg sign (gluteus medius/minimus weakness)

4. MUSCULATURE OF THE LOWER LIMB



Reference image — supplementary visual aid, INDIGENE Study Console

Deep Group (Lateral Rotators)

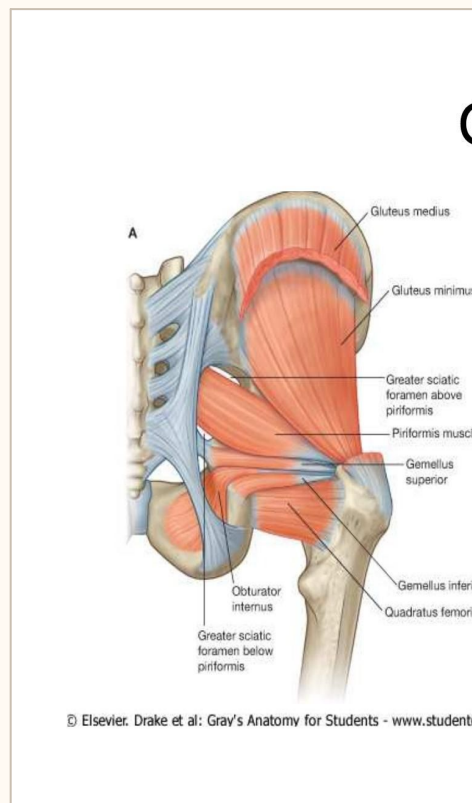
Muscle	Origin	Insertion	Action	Nerve
Piriformis	Anterior sacrum, just lateral to the anterior sacral foramina	Upper surface of the greater trochanter	Lateral rotation (extended hip); abduction (flexed hip)	Sacral nerves S1-S2
Gemellus superior	Lateral ischial spine, above the ischial tuberosity	Common tendon with obturator internus + gemellus inferior → posteromedial greater trochanter	Lateral rotation (extended hip); abduction (flexed hip)	Obturator internus nerve
Gemellus inferior	Upper ischial tuberosity, below its upper margin	Same common tendon as above	Lateral rotation (extended hip); abduction (flexed hip)	Quadratus femoris nerve
Obturator internus	Inner margins of the obturator foramen + obturator membrane	Same common tendon → posteromedial greater trochanter	Lateral rotation (extended hip); abduction (flexed hip)	Obturator internus nerve
Quadratus femoris	Lateral ischium, anterior to the ischial tuberosity	Quadratus tubercle of the femur, medial to the intertrochanteric crest	External (lateral) rotation of the femur	Quadratus femoris nerve

EXAM HACK

Three of the deep six (obturator internus, gemellus superior, gemellus inferior) converge on ONE common tendon at the posteromedial greater trochanter. Quadratus femoris is the exception — its own dedicated insertion at the quadratus tubercle, and its own dedicated nerve (which also supplies gemellus inferior). Piriformis passes through the greater sciatic foramen — its relationship to the sciatic nerve (usually below it, sometimes through/above it) is an anatomical variant that underlies piriformis syndrome.

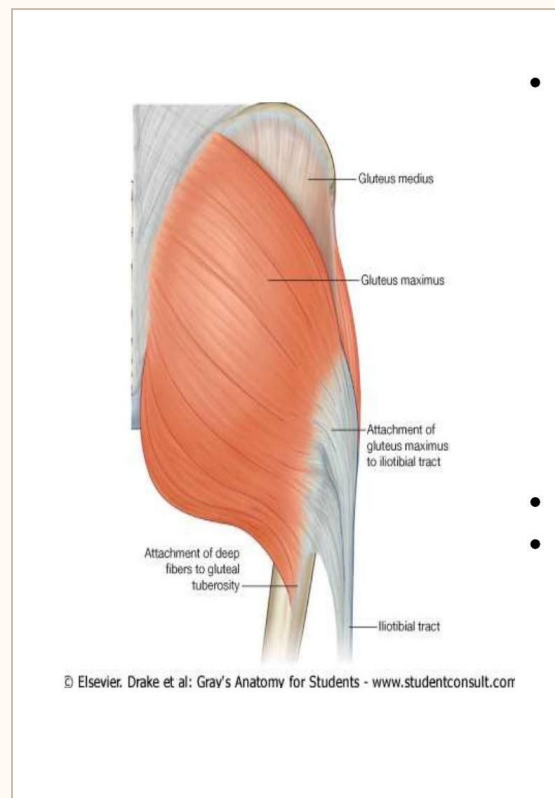
Deep lateral rotators — piriformis, obturator internus, gemelli, quadratus femoris (with greater/lesser sciatic foramen)

4. MUSCULATURE OF THE LOWER LIMB



Reference image — supplementary visual aid, INDIGENE Study Console

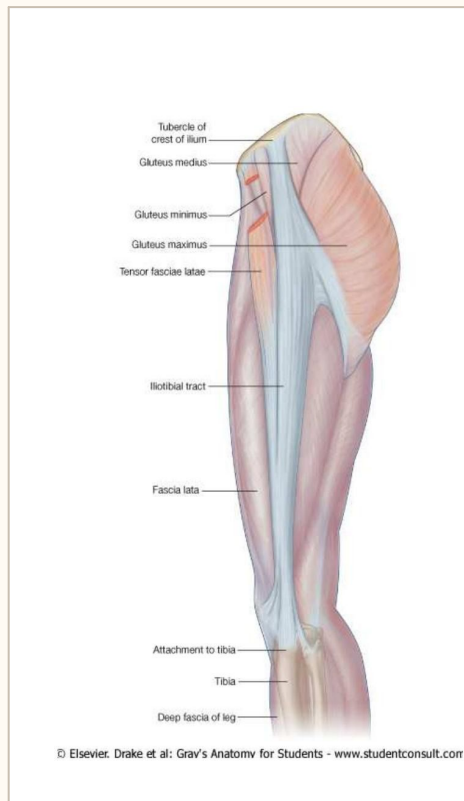
Gluteus maximus — origin, fibre direction, and insertion into the iliotibial tract



Reference image — supplementary visual aid, INDIGENE Study Console

Tensor fasciae latae and the iliotibial tract — full lateral thigh/leg view

4. MUSCULATURE OF THE LOWER LIMB



Reference image — supplementary visual aid, INDIGENE Study Console

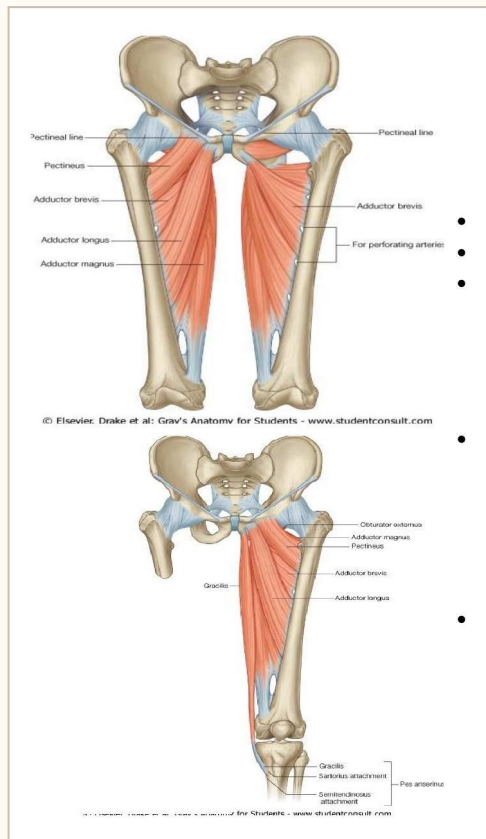
Muscles of the Thigh

Medial Compartment

Muscle	Origin	Insertion	Action	Nerve
Obturator externus	External margin of the obturator foramen + obturator membrane	Trochanteric fossa (passing posterior to the femoral neck)	Lateral rotation of the thigh	Obturator nerve
Gracilis	Medial + inferior surfaces of the ventral pubic bone (ischiopubic ramus)	Medial proximal tibia at the level of the tibial tuberosity (pes anserinus)	Flexion, adduction, medial rotation of the femur (+ assists knee flexion)	Obturator nerve
Pectineus	Pectineal line of the pubis	Pectineal line on the posterior femoral shaft	Flexion, adduction, internal rotation of the femur	Femoral nerve
Adductor brevis	Ventral surface of the pubic bone	Upper linea aspera	Flexion, adduction, internal rotation	Obturator nerve
Adductor longus	Superior ventral pubic bone, below the pubic crest	Middle portion of the linea aspera	Flexion, adduction, internal rotation	Obturator nerve
Adductor magnus	Ischiopubic ramus (adductor part) + ischial tuberosity (hamstring part)	Adductor part: quadrate tubercle base, gluteal tuberosity, lower linea aspera. Hamstring part: adductor tubercle	Extension, flexion, adduction, medial rotation	Obturator nerve (adductor part) + Tibial nerve (hamstring part)

Adductor group, pectineus and gracilis — superficial and deep layers

4. MUSCULATURE OF THE LOWER LIMB



Reference image — supplementary visual aid, INDIGENE Study Console

EXAM HACK

Pectineus is femoral nerve despite sitting in the medial compartment — the classic "border-crosser" exception. Adductor magnus is the split-innervation muscle here — obturator nerve for the adductor part, tibial nerve for the hamstring part — directly paralleling brachialis and FDP's dual-innervation pattern in the upper limb.

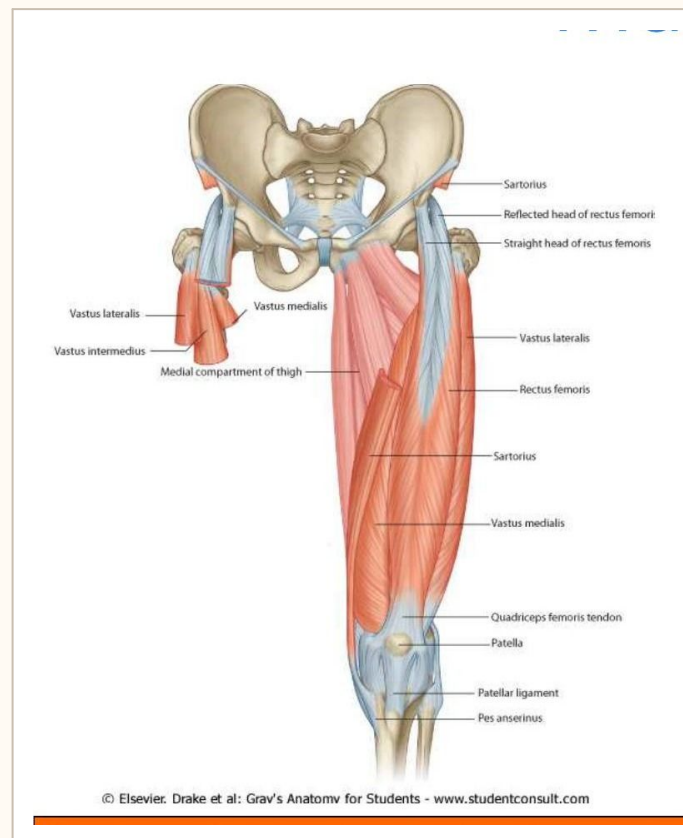
Anterior Compartment

Muscle	Origin	Insertion	Action	Nerve
Psoas major	Transverse processes + vertebral bodies of T12-L4	Lesser trochanter	Flexion + lateral rotation of the femur	Anterior branches of L1-L3
Iliacus	Upper iliac fossa, below the iliac crest	Lesser trochanter (joins psoas major as "iliopsoas")	Flexion + internal rotation of the femur	Femoral nerve
Sartorius	ASIS	Medial proximal tibia at the tibial tuberosity level (pes anserinus)	Flexion, abduction, external rotation of the femur + flexion of the leg	Femoral nerve
Rectus femoris	A groove on the dorsal ilium above the acetabulum + AIIS	Quadriceps femoris tendon	Flexion of the thigh + extension of the leg	Femoral nerve
Vastus lateralis	Inferior margin of the greater trochanter, lateral intertrochanteric line, lateral linea aspera	Lateral quadriceps tendon	Extension of the leg	Femoral nerve
Vastus medialis	Medial intertrochanteric line, pectineal line, medial linea aspera	Medial quadriceps tendon	Extension of the leg	Femoral nerve
Vastus intermedius	Anterior + lateral femoral shaft (posterior to rectus femoris)	Quadriceps femoris tendon	Extension of the leg	Femoral nerve

Vastus lateralis, rectus femoris, vastus intermedius, and vastus medialis are collectively the **quadriceps femoris**.

Quadriceps femoris (all 4 heads) and sartorius — all femoral-nerve innervated

4. MUSCULATURE OF THE LOWER LIMB



Reference image — supplementary visual aid, INDIGENE Study Console

EXAM HACK

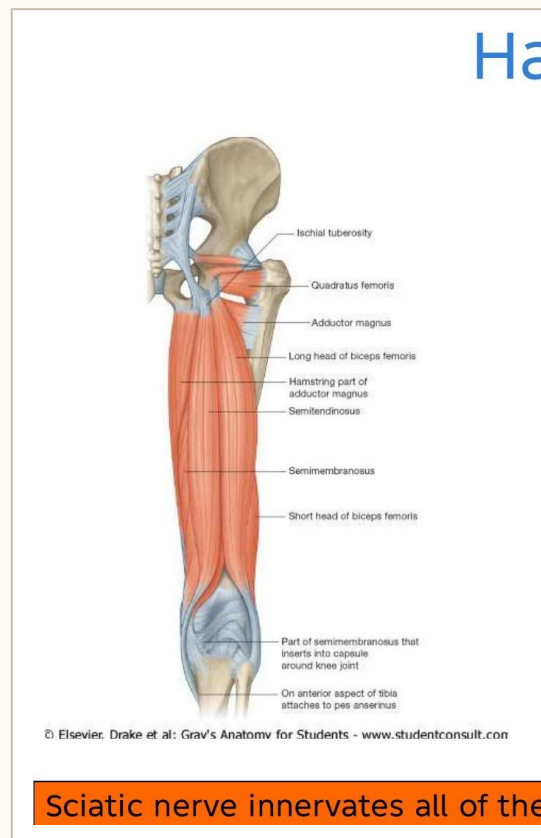
Psoas major (L1-L3 direct nerve branches) and iliacus (femoral nerve) are separate muscles with separate innervation, but they insert together at the lesser trochanter, functioning jointly as "iliopsoas" — the primary hip flexor. Rectus femoris is the only quad muscle crossing BOTH hip and knee, letting it flex the thigh AND extend the leg simultaneously (e.g. kicking).

Posterior Compartment (Hamstrings)

Muscle	Origin	Insertion	Action	Nerve
Semimembranosus	Superolateral dorsal ischial tuberosity	Groove on the medial tibial condyle	Extension + internal rotation of the thigh; flexion + internal rotation of the leg	Sciatic nerve (tibial division)
Semitendinosus	Inferomedial dorsal ischial tuberosity	Medial tibia at the tibial tuberosity level (pes anserinus)	Extension + internal rotation of the thigh; flexion + internal rotation of the leg	Sciatic nerve (tibial division)
Biceps femoris — long head	Superomedial ischial tuberosity	Lateral head of the fibula	Extension + external rotation of the thigh; flexion + external rotation of the leg	Sciatic nerve (tibial division)
Biceps femoris — short head	Lateral mid-lower linea aspera	Lateral head of the fibula (joins long head)	Flexion + external rotation of the leg	Sciatic nerve (common peroneal division)

Hamstrings — biceps femoris (both heads), semitendinosus, semimembranosus

4. MUSCULATURE OF THE LOWER LIMB



Reference image — supplementary visual aid, INDIGENE Study Console

EXAM HACK

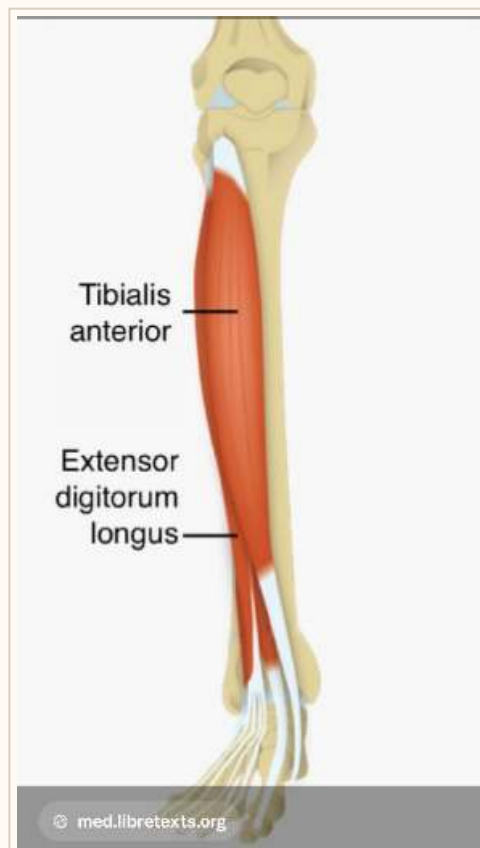
Biceps femoris short head is the single most tested hamstring exception: it doesn't cross the hip (origin is the femur itself, not the ischial tuberosity), so it's supplied by the common peroneal division of the sciatic nerve — every other hamstring gets the tibial division. Pes anserinus = sartorius (femoral n.) + gracilis (obturator n.) + semitendinosus (tibial n.) converging at one point on the medial tibia — "Say Grace before Tea."

MUSCULATURE — LEG, ANTERIOR COMPARTMENT (Reference Images)

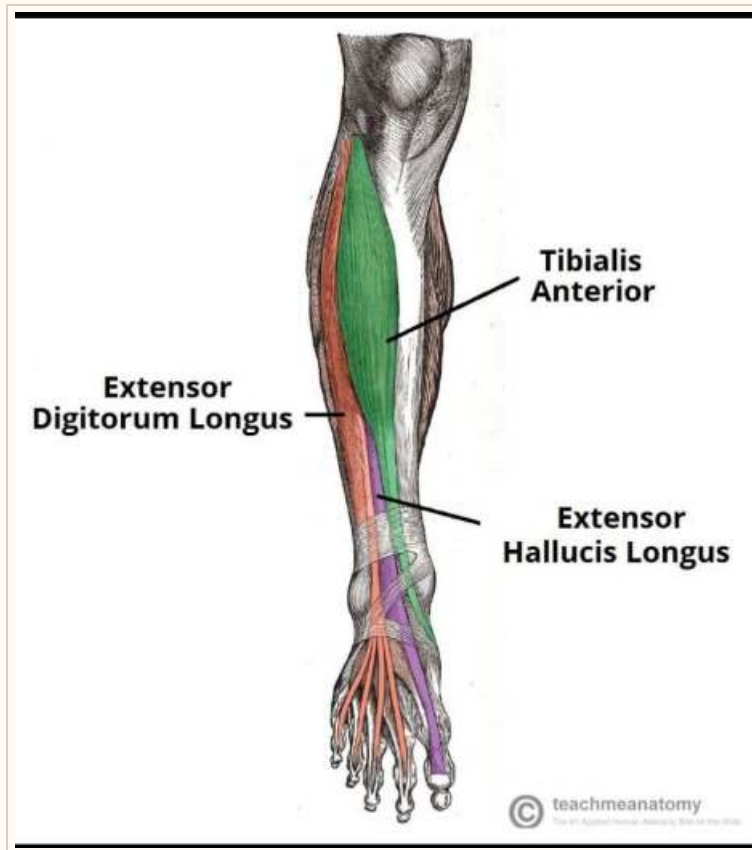
Tibialis anterior, EDL, patella/patellar ligament, soleus



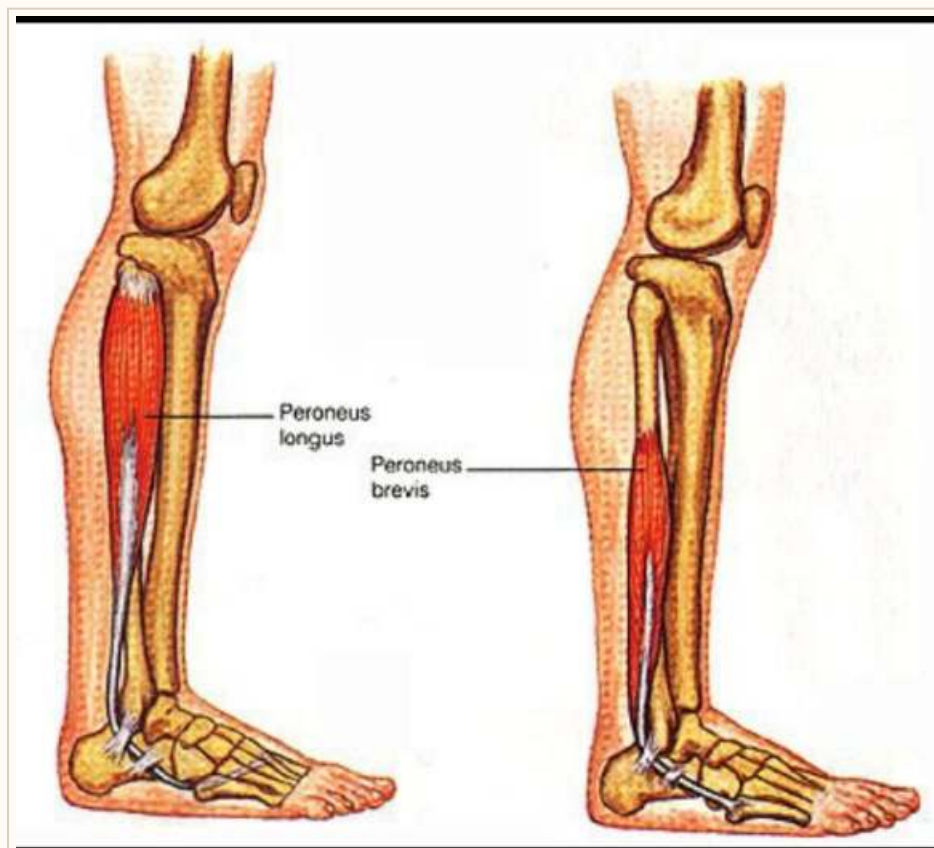
Tibialis anterior & extensor digitorum longus



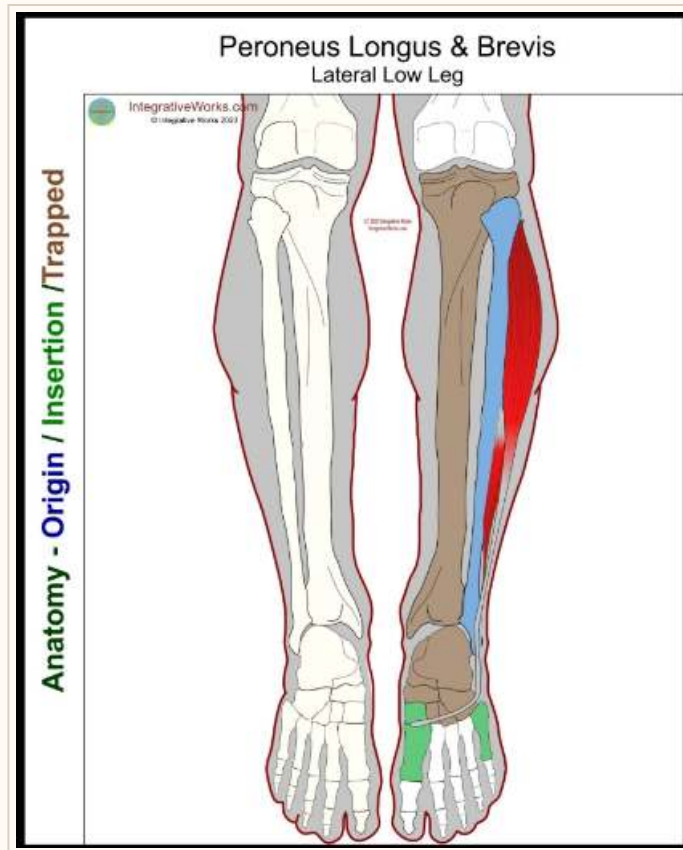
Tibialis anterior, EDL, extensor hallucis longus



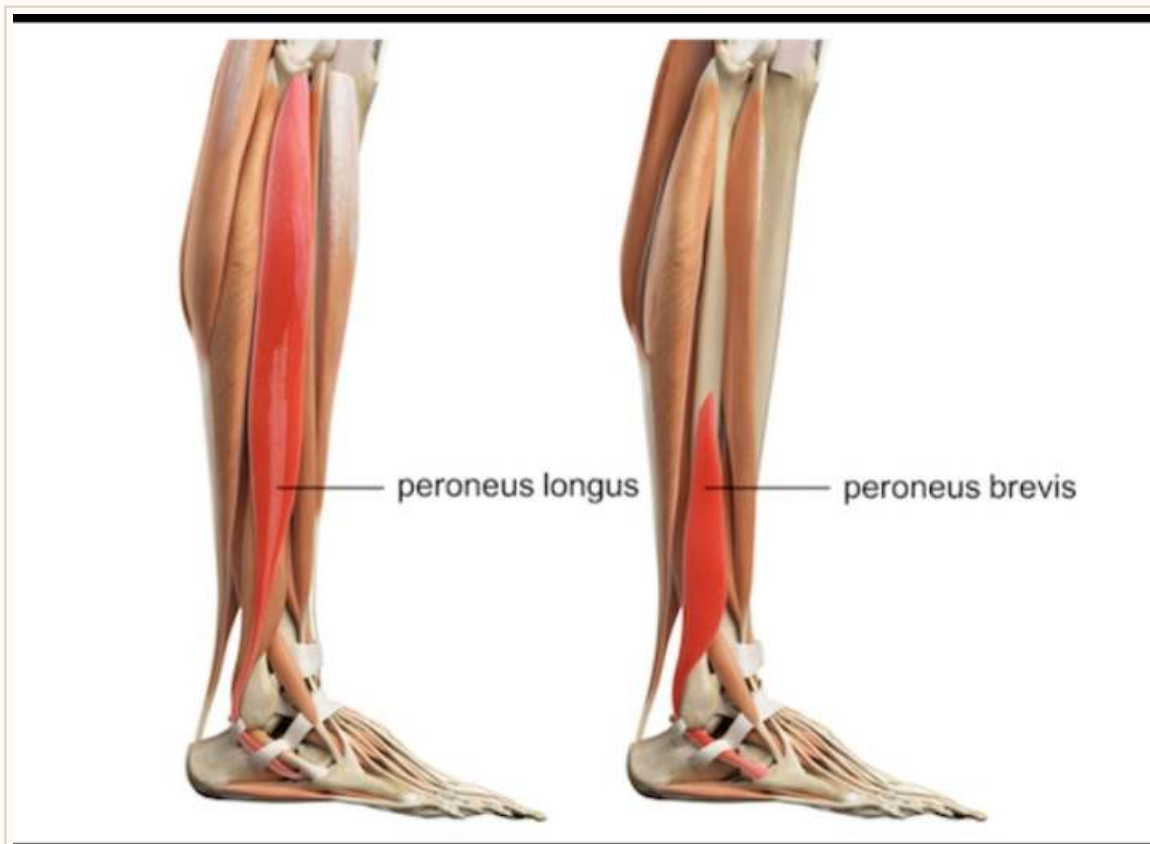
Peroneus (fibularis) longus & brevis



Peroneus longus & brevis — origin/insertion



Peroneus longus & brevis — lateral leg views



ADDED DETAIL

Peroneus tertius is functionally/developmentally a distal slip of extensor digitorum longus — hence its deep peroneal nerve supply, unlike peroneus longus/brevis which are superficial peroneal territory.

Deep Posterior Compartment

Muscle	Origin	Insertion	Action	Nerve
Popliteus	Lateral surface of the lateral femoral condyle	Posteromedial proximal tibia	Laterally rotates the femur to "unlock" the knee from full extension	Tibial nerve
Flexor hallucis longus	Posterior + medial mid-lower fibular shaft	Plantar surface of the great toe's distal phalanx	Plantarflexion of the foot + flexion of the great toe	Tibial nerve
Tibialis posterior	Posterolateral tibia (lateral to the soleal line), posterior interosseous membrane, posteromedial fibula	Plantar surface of the medial cuneiform	Plantarflexion + inversion of the foot; dynamic support of the medial longitudinal arch	Tibial nerve
Flexor digitorum longus	Posteromedial tibial shaft	Plantar surface of the distal phalanges of the lateral 4 toes (4 tendons)	Plantarflexion of the foot + flexion of the lateral 4 toes	Tibial nerve

Superficial Posterior Compartment

Muscle	Origin	Insertion	Action	Nerve
Plantaris	Lateral supracondylar line of the distal femur	Achilles (calcaneal) tendon	Flexion of the leg + plantarflexion of the foot	Tibial nerve
Soleus	Soleal line + posterior fibular head	Achilles (calcaneal) tendon	Plantarflexion of the foot	Tibial nerve
Gastrocnemius	2 heads: posterosuperior lateral femoral condyle + posterosuperior medial femoral condyle	Achilles (calcaneal) tendon	Flexion of the leg + plantarflexion of the foot	Tibial nerve

EXAM HACK — "TOM, DICK, AND HARRY"

Behind the medial malleolus, anterior to posterior: **T**ibialis posterior, flexor **D**igitorum longus, posterior tibial **A**rtery/vein, tibial **N**erve (the "AND"), flexor **H**allucis longus. Compression here = tarsal tunnel syndrome, the lower limb's carpal tunnel equivalent. Popliteus "unlocks" the knee — the muscle that initiates flexion from full extension, reversing the screw-home mechanism. Gastrocnemius crosses the knee AND ankle (fast, powerful); soleus crosses only the ankle (postural, endurance) — test soleus alone with the knee flexed.

Muscles of the Foot (Intrinsic)

Alzheimer's Mode: Dorsal muscles (2) + 4 plantar layers, mirroring the hand's intrinsic layering but with foot-specific additions like quadratus plantae.

Dorsal Muscles

Muscle	Origin	Insertion	Action	Nerve
Extensor hallucis brevis	Superolateral calcaneus	Proximal phalanx of the great toe	Extension of the great toe	Deep peroneal nerve
Extensor digitorum brevis	Superolateral calcaneus	Proximal phalanges of the lateral 4 toes (4 tendons)	Extension of the lateral 4 toes	Deep peroneal nerve

1st Plantar Layer

Muscle	Origin	Insertion	Action	Nerve
Abductor hallucis	Medial process of the calcaneal tuberosity	Medial proximal phalanx of the great toe	Abduction + flexion of the great toe	Medial plantar nerve
Flexor digitorum brevis	Medial process/notch of the calcaneal tuberosity	Proximal phalanges of the lateral 4 toes (4 tendons)	Flexion of the lateral 4 toes	Medial plantar nerve
Abductor digiti minimi	Lateral process of the calcaneal tubercle	Lateral proximal phalanx of the little toe	Abduction of the little toe	Lateral plantar nerve

2nd Plantar Layer

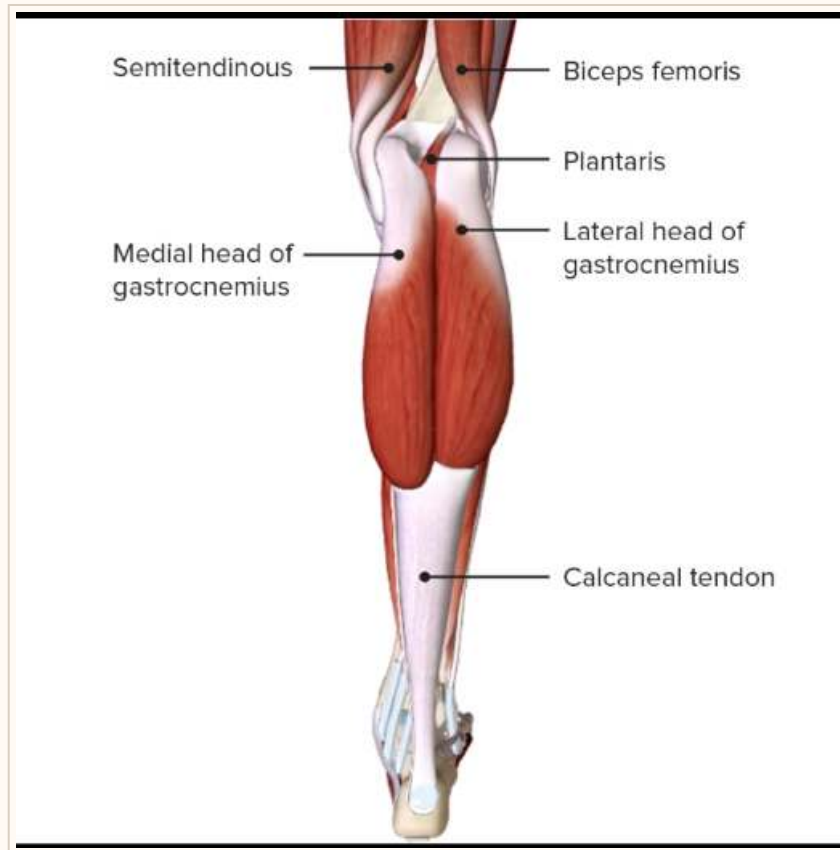
Muscle	Origin	Insertion	Action	Nerve
Quadratus plantae	Medial + lateral processes of the calcaneal tuberosity	Lateral margin of the flexor digitorum longus tendon	Flexion of the lateral 4 toes (corrects FDL's oblique pull)	Lateral plantar nerve
Lumbricals (4)	Adjacent borders of the flexor digitorum longus tendons	Medial surface of the proximal phalanges (toes 2-5)	Extension of the phalanges + flexion of the lateral 4 toes	2 medial lumbricals: medial plantar nerve. 2 lateral lumbricals: lateral plantar nerve

CORRECTED — LUMBRICAL NERVE SPLIT IS 2:2, NOT 1:3

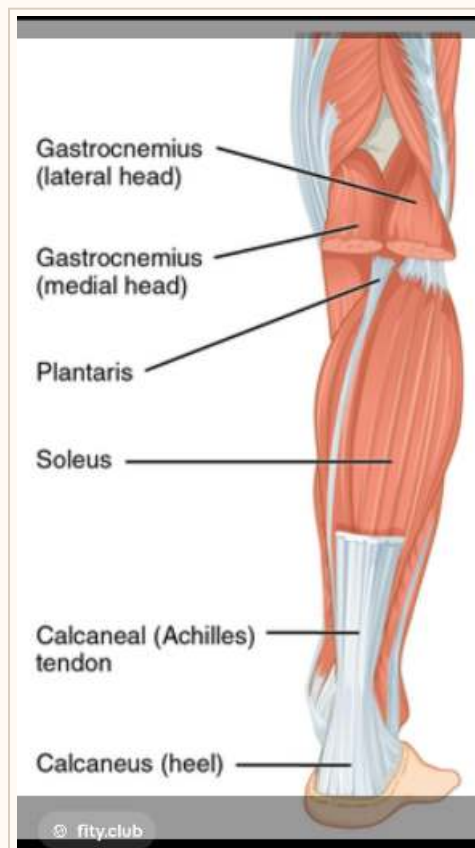
Per your uploaded course material: the TWO medial lumbricals are medial plantar nerve, and the TWO lateral lumbricals are lateral plantar nerve — a 2:2 split. An earlier version of this teaching (and an earlier draft of this section) stated a 1:3 split (only the most medial lumbrical = medial plantar nerve), which is how some other anatomy texts describe it, but it does not match your own course material. Go with the 2:2 split your lecturer is teaching — that's what

MUSCULATURE — LEG, SUPERFICIAL POSTERIOR COMPARTMENT (Reference

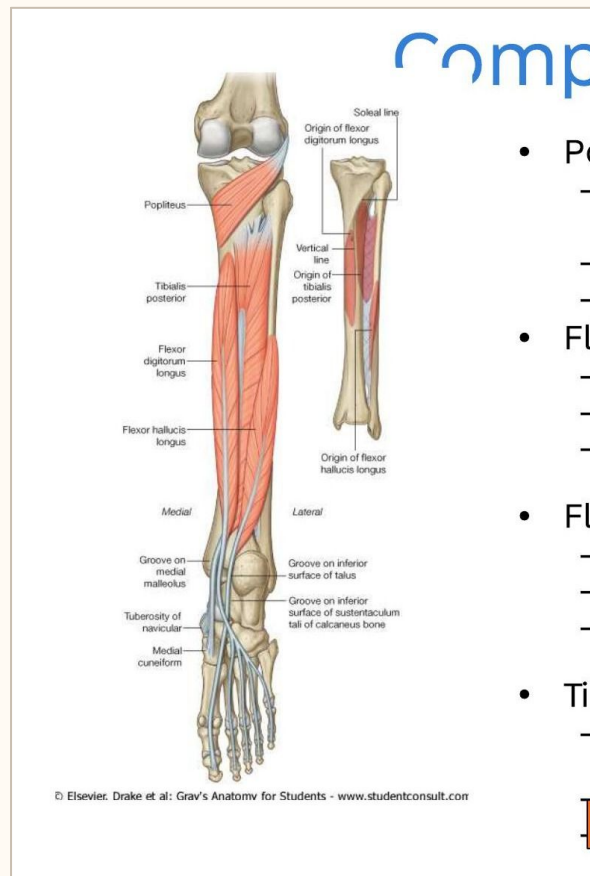
Semitendinosus, biceps femoris, plantaris, gastrocnemius, calcaneal tendon



Gastrocnemius heads, plantaris, soleus, calcaneal tendon, calcaneus



Deep posterior compartment — popliteus, flexor digitorum longus, flexor hallucis longus, tibialis posterior



Reference image — supplementary visual aid, INDIGENE Study Console

you'll be examined on.

3rd Plantar Layer

Muscle	Origin	Insertion	Action	Nerve
Flexor hallucis brevis	Cuboid + lateral cuneiform	Medial + lateral proximal phalanx of the great toe	Flexion of the great toe	Medial plantar nerve
Adductor hallucis — oblique head	Proximal 3rd + 4th metatarsals	Lateral proximal phalanx of the great toe	Adduction of the great toe	Lateral plantar nerve
Adductor hallucis — transverse head	Ligaments around the 3rd-5th metatarsophalangeal joints	Lateral proximal phalanx of the great toe	Adduction of the great toe	Lateral plantar nerve
Flexor digiti minimi brevis	Proximal 5th metatarsal	Lateral proximal phalanx of the little toe	Flexion of the little toe	Lateral plantar nerve

4th Plantar Layer

Muscle	Origin	Insertion	Action	Nerve
Plantar interossei (3)	Medial surface of the 3rd, 4th, 5th metatarsals	Proximal phalanx bases (same toes)	Adduction of the lateral 3 toes	Lateral plantar nerve
Dorsal interossei (4)	Lateral surfaces of the 2nd-4th metatarsals (1st interosseus also from medial 2nd + lateral 1st metatarsal)	Medial side of the proximal phalanx bases (adjacent digits)	Abduction of the 2nd, 3rd, 4th toes	2 medial: deep peroneal nerve. 2 lateral: lateral plantar nerve

EXAM HACK

Foot interossei nerve supply has a twist versus the hand: dorsal interossei get SPLIT innervation (2 medial = deep peroneal nerve, 2 lateral = lateral plantar nerve) — unlike the hand where ALL interossei are uniformly ulnar nerve. Quadratus plantae has no hand equivalent — it exists purely to correct FDL's oblique pull into a straight posterior-to-anterior line.

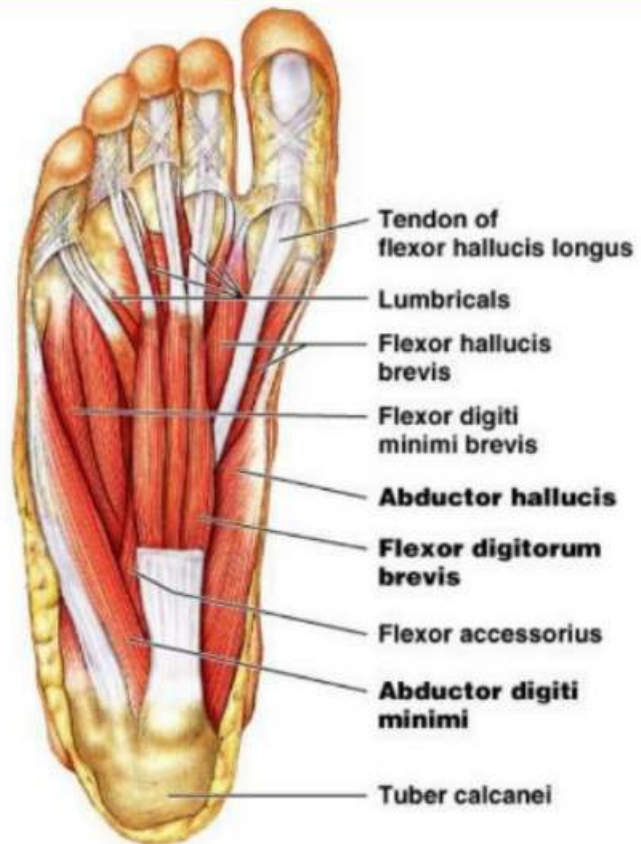
CLINICAL

Tarsal tunnel syndrome can affect both medial and lateral plantar nerves since both branch from the tibial nerve at that exact site. Claw toes/hammer toes result from intrinsic muscle imbalance, mirroring claw hand from lumbrical loss in the upper limb.

Plantar muscles — first (superficial) layer

Plantar Muscles: First Layer (Superficial)

- Superficial muscles of the plantar aspect of the foot
- These muscles are similar to the corresponding muscles of the hand



(a) First layer

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Figure 10.25a

Deep & superficial plantar views — all 4 layers

Intrinsic Muscles of the Foot

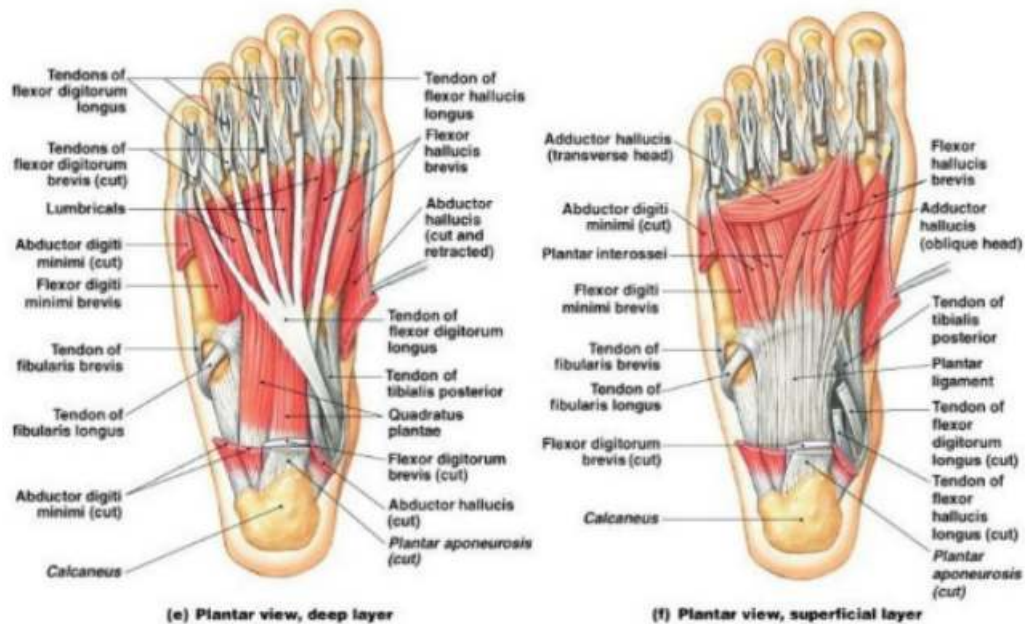


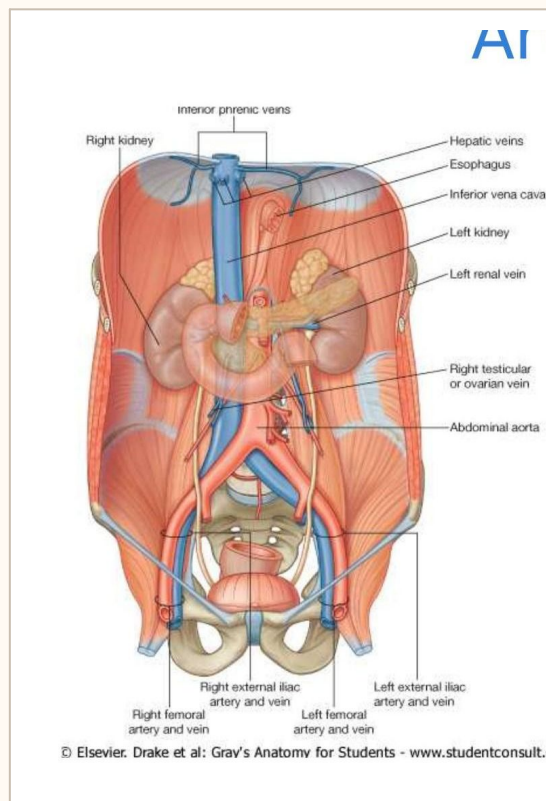
Figure 11.18e Deep Foot

Figure 11.18f Superficial Plantar View

5. VASCULATURE OF THE LOWER LIMB

The arterial supply begins as the **external iliac artery**, which becomes the **femoral artery** once it passes beneath the inguinal ligament. The chain continues: femoral → popliteal (via the adductor hiatus) → anterior tibial + posterior tibial → dorsalis pedis + plantar arteries.

Abdominal aorta → common iliac → internal/external iliac → femoral artery



Reference image — supplementary visual aid, INDIGENE Study Console

Femoral Artery — Branches Just Below the Inguinal Ligament

Alzheimer's Mode: Immediately after becoming the femoral artery, 5 small branches fan out before the "real" workhorse branch (*profunda femoris*) takes over supplying the thigh.

Branch	Course	Supplies
Superficial circumflex iliac artery	Antero-lateral surface, ascends	Skin of the anterolateral abdomen/groin + superficial inguinal lymph nodes
Superficial epigastric artery	Anterior surface, ascends	Skin + muscles of the abdominal wall
Superficial external pudendal artery	Anteromedial surface, runs inferomedially	Pelvic muscles + scrotal/labial (mostly labia majus) skin
Deep external pudendal artery	Posteromedial surface, runs medially	Pelvic muscles + scrotal/labial skin
Profunda femoris artery (deep femoral artery)	Posterior surface, descends laterally through the deep thigh compartments	The main supplier of the thigh — see below

ADDED DETAIL

The four superficial branches (circumflex iliac, epigastric, and both external pudendal arteries) arising immediately below the inguinal ligament were not part of the earlier teaching, which jumped straight to profunda femoris. These four are small but genuinely examinable — they supply the groin/abdominal wall skin and lymphatics before the profunda femoris takes over the heavy lifting.

Profunda Femoris and Its Branches

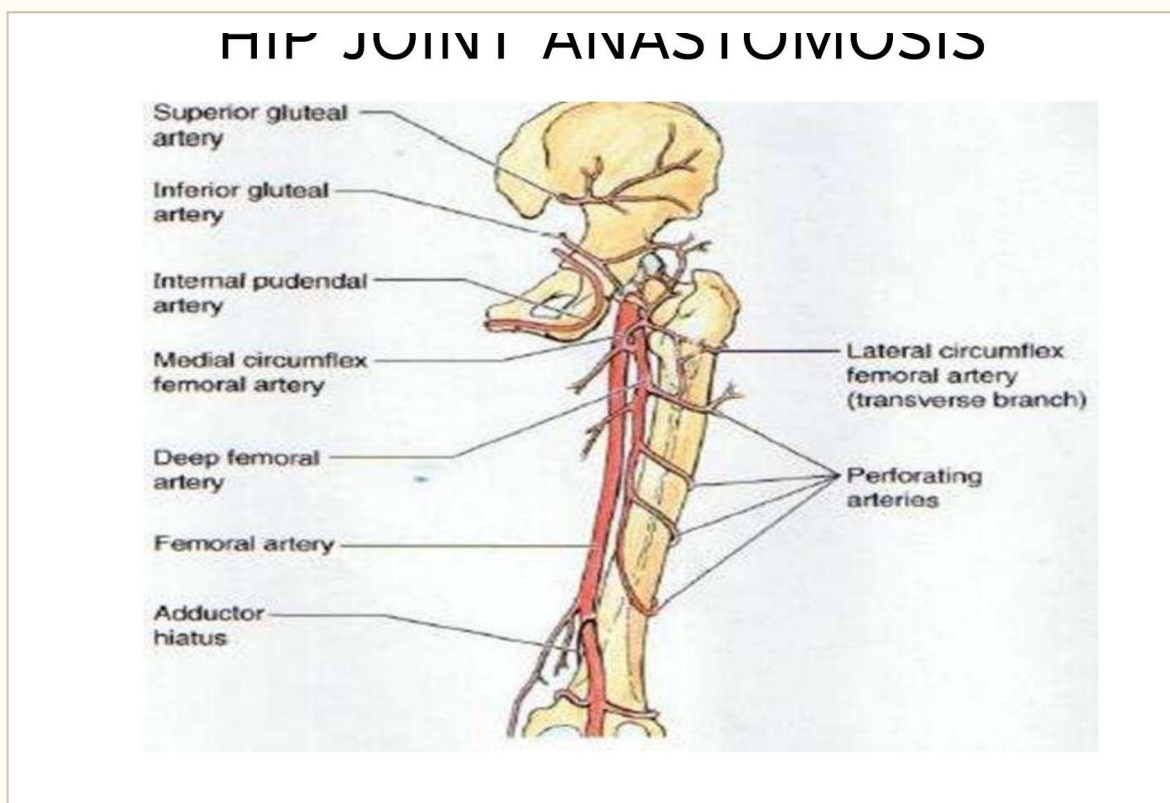
5. VASCULATURE OF THE LOWER LIMB

Branch	Course	Supplies
Medial circumflex femoral artery	Medial surface, loops around psoas major into the posterior thigh	Muscles, head + neck of femur; ends anastomosing with the lateral circumflex femoral artery
Lateral circumflex femoral artery	Lateral surface, gives 3 branches: descending (to the knee, anastomoses with popliteal branches), ascending (to the neck/head of femur, anastomoses with the medial circumflex femoral), transverse (loops around the femur to the cruciate anastomosis)	Adjacent skin/knee, neck + head of femur, proximal femoral anastomotic network
Perforating arteries (4)	Arise at different levels from the lateral surface, pierce adductor magnus's tendon to reach the posterior thigh, giving ascending/descending branches connecting to one another	Posterior thigh (hamstrings); contribute to anastomoses at both the hip (above) and knee (below)

EXAM HACK

The "cruciate anastomosis" at the proximal femur is formed by the medial circumflex femoral artery, lateral circumflex femoral artery, superior gluteal artery, and inferior gluteal artery — a 4-way anastomotic network around the femoral neck/head, directly relevant to why femoral neck fractures don't always cause complete avascular necrosis (some collateral flow may persist).

Hip joint (cruciate/trochanteric) anastomosis around the femoral neck — every named branch labeled



Reference image — supplementary visual aid, INDIGENE Study Console

Obturator Artery

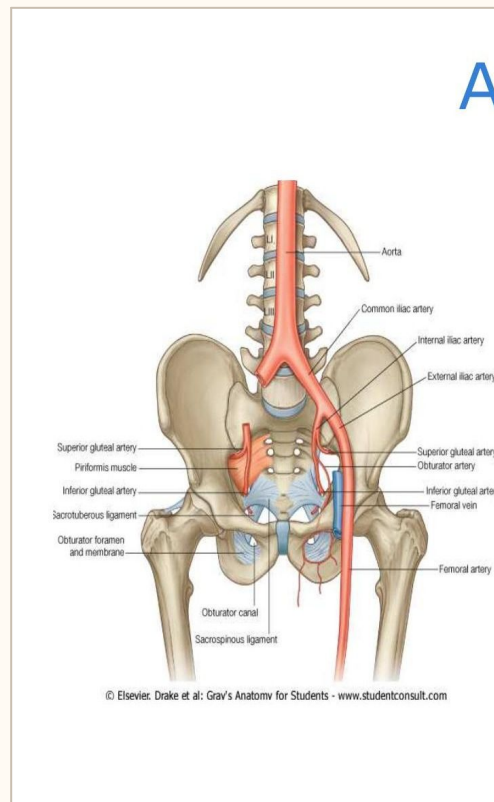
Arises from the anterior division of the internal iliac artery in the pelvis, running inferolaterally and dividing into an **anterior branch** (through the obturator canal into the medial thigh, supplying the medial thigh muscles/skin) and a **posterior branch** (supplies the acetabulum at the hip joint).

ADDED DETAIL

This vessel, distinct from the obturator nerve already covered, directly supplies the medial thigh alongside the obturator nerve — and its posterior branch is a contributor to acetabular/hip joint blood supply, relevant background for the femoral head circulation discussion in the hip joint section.

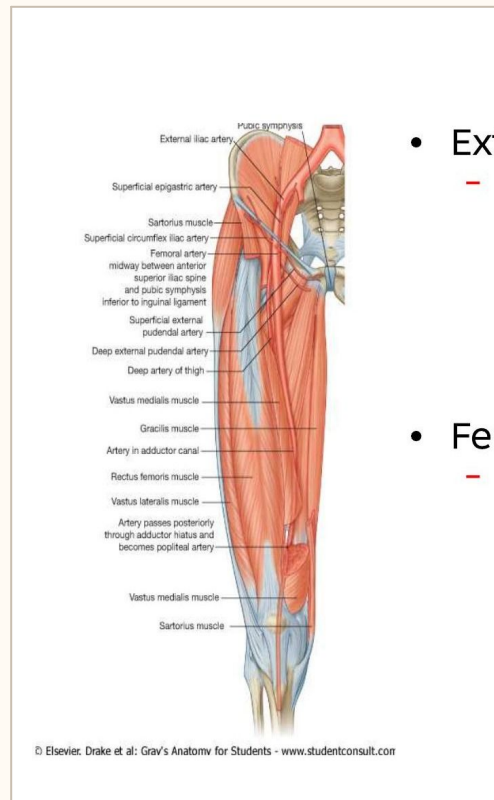
5. VASCULATURE OF THE LOWER LIMB

Internal iliac branches — superior/inferior gluteal, internal pudendal and obturator arteries relative to piriformis



Reference image — supplementary visual aid, INDIGENE Study Console

Femoral artery → deep femoral → through adductor hiatus → becomes the popliteal artery



Reference image — supplementary visual aid, INDIGENE Study Console

Popliteal Artery and the Genicular Anastomosis

5. VASCULATURE OF THE LOWER LIMB

At the adductor hiatus (inferior margin of adductor magnus), the femoral artery becomes the **popliteal artery**, entering the posterior thigh/knee region. It is the deepest structure in the popliteal fossa.

Alzheimer's Mode: Around the knee, SIX named arteries weave together into one dense anastomotic net — this collateral network is exactly why the knee can often survive a popliteal artery injury better than you'd expect.

Artery	Origin	Connects to
Descending genicular artery	Anterior femoral artery, just above the adductor hiatus	Superior medial genicular artery (above the knee) + inferior medial genicular artery (below the knee)
Superior medial genicular artery	Medial popliteal artery, above the epicondyles	Inferior medial genicular artery
Superior lateral genicular artery	Lateral popliteal artery, above the femoral epicondyles	Descending branch of the lateral circumflex femoral artery + inferior lateral genicular artery
Inferior medial genicular artery	Medial popliteal artery, at the level of the condyles	All the above branches
Inferior lateral genicular artery	Lateral popliteal artery	All the above + circumflex fibular artery + recurrent anterior tibial artery
Circumflex fibular artery	Posterior tibial artery, at the tibial tuberosity level	Circles the fibular neck, connects to the inferior lateral genicular + recurrent anterior tibial arteries
Recurrent anterior tibial artery	Anteromedial anterior tibial artery	Circumflex fibular + inferior + superior lateral genicular arteries

ADDED DETAIL

Earlier teaching noted "genicular branches form an anastomotic network" in passing. This is the actual named 6-7 vessel network in full — a frequently tested structure, especially regarding collateral circulation if the popliteal artery is occluded or injured (e.g., in a knee dislocation).

Anterior Tibial Artery and Branches

The popliteal artery splits below the knee into anterior and posterior tibial arteries. The **anterior tibial artery** arises laterally, enters the anterior leg compartment through the superior opening in the interosseous membrane, and descends along its anterior surface.

Branch	Course/Supplies
Recurrent anterior tibial artery	Ascends to connect with the circumflex fibular + genicular arteries (see above)
Recurrent posterior tibial artery	Arises before the artery enters the anterior compartment, ascends to anastomose with the inferior lateral genicular artery
Muscular branches	Numerous, supply the muscles + skin of the anterior leg
Anterior medial malleolar artery	Curves around the medial malleolus, supplies the medial ankle + overlying skin
Anterior lateral malleolar artery	Curves around the lateral malleolus, supplies the lateral ankle + overlying skin
Dorsalis pedis artery	Terminal continuation of the anterior tibial artery onto the dorsum of the foot

Dorsalis Pedis Artery — Branches

Branch	Course/Supplies
Medial tarsal artery	Medial surface, at the anterior navicular level, supplies the medial foot dorsum

5. VASCULATURE OF THE LOWER LIMB

Branch	Course/Supplies
Lateral tarsal artery	Lateral surface, at the posterior navicular level; anastomoses with the anterior lateral malleolar artery, supplies the lateral foot dorsum
Arcuate artery	Lateral surface, at the anterior intermediate cuneiform level; runs over the lateral 4 metatarsals, giving off the 2nd, 3rd, and 4th metatarsal arteries (each splitting into two terminal digital branches)
Deep plantar artery	Runs deep into the foot to join the plantar arch
First dorsal metatarsal artery	Gives two branches running along the sides of the great toe

Posterior Tibial Artery and Branches

The direct continuation of the popliteal artery into the posterior leg compartment, providing the major blood supply there.

Branch	Course/Supplies
Circumflex fibular artery	Circles the fibular neck (see genicular network above)
Fibular artery	Courses along the lateral leg, supplying the lateral + posterior compartment muscles/skin
Muscular branches	Numerous, supply the posterior compartment muscles + skin
Medial plantar artery	Runs along the medial sole to the medial great toe, supplying the medial sole
Lateral plantar artery	Runs along the lateral sole, gives a small branch to the lateral little toe, then crosses medially to form the plantar arch

Plantar Arch — Branches

The plantar arch (formed by the lateral plantar artery, completed by the deep plantar artery from dorsalis pedis) gives off 4 **plantar metatarsal arteries**, running through the metatarsal spaces and each dividing into two terminal digital branches along the sides of adjacent phalanges.

EXAM HACK

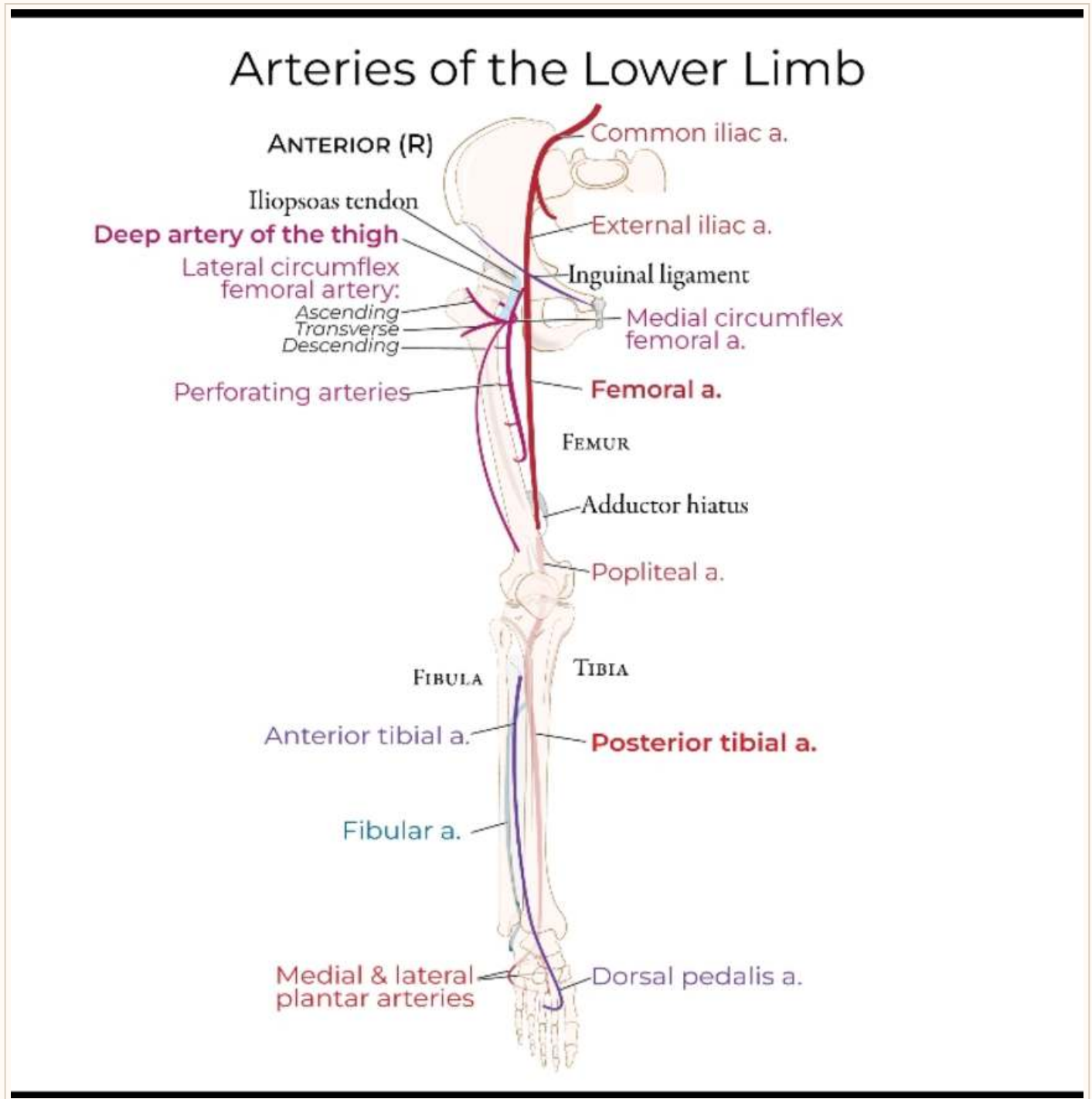
Anterior tibial artery (→ dorsalis pedis) supplies the DORSUM of the foot; posterior tibial artery (→ medial/lateral plantar arteries) supplies the SOLE. The plantar arch is completed by BOTH systems meeting — deep plantar artery (from dorsalis pedis, anterior system) joins the lateral plantar artery (posterior system) — exactly mirroring how the hand's deep palmar arch is radial-dominant with an ulnar contribution.

CLINICAL

Peripheral arterial disease is assessed via the dorsalis pedis pulse (dorsum, between the 1st/2nd metatarsals) and the posterior tibial pulse (behind the medial malleolus). Popliteal artery aneurysm is the most common peripheral artery aneurysm. Compartment syndrome risks the anterior tibial artery specifically, given the tight anterior compartment fascia.

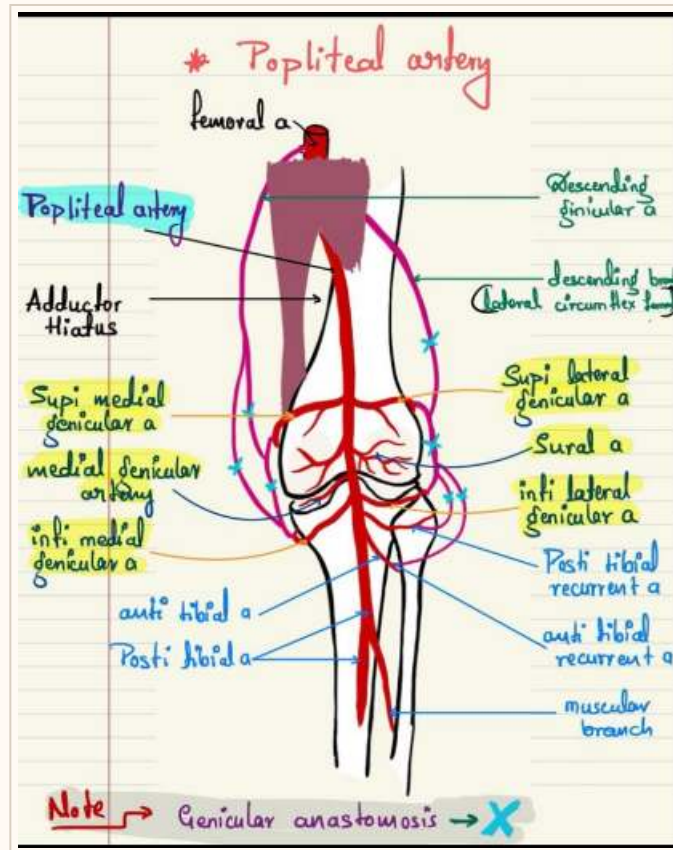
VASCULATURE — ARTERIES OF THE LOWER LIMB (Reference Image)

Femoral → popliteal → tibial → dorsalis pedis / plantar arteries

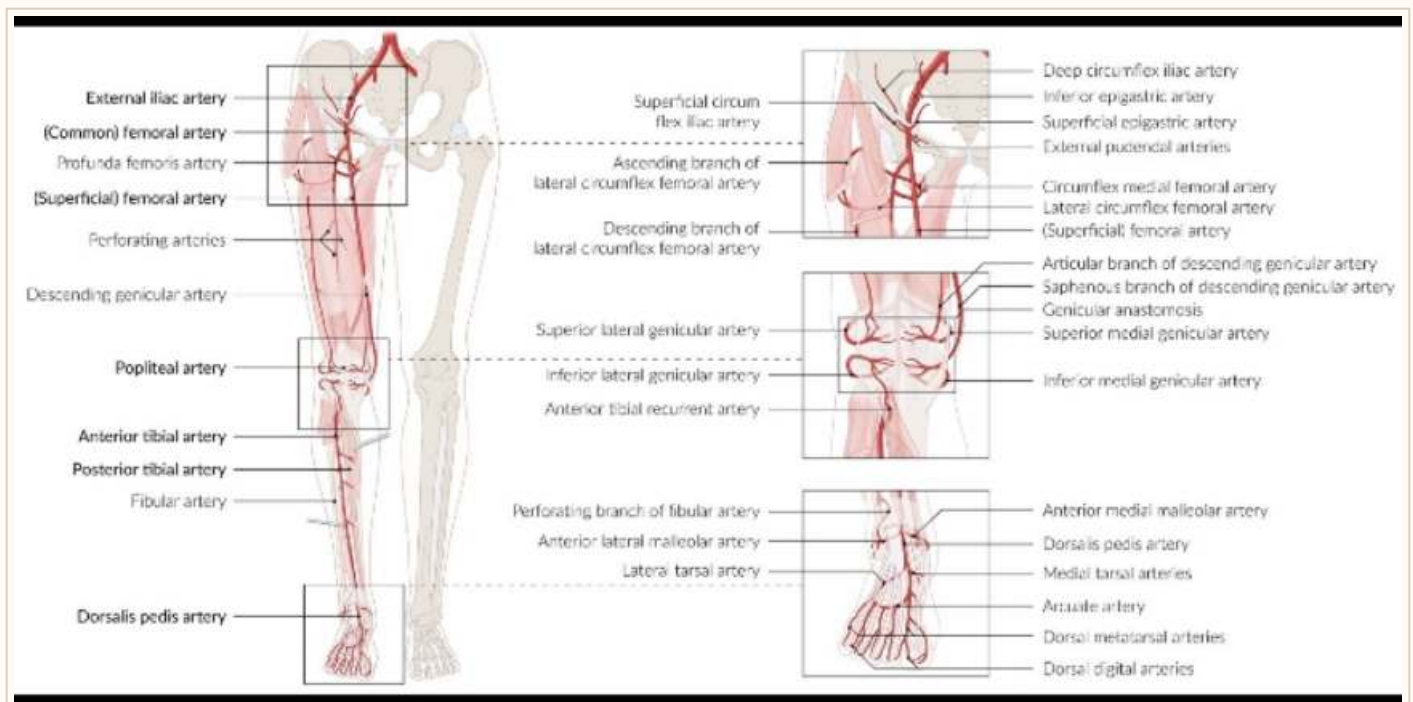


VASCULATURE — GENICULAR ANASTOMOSIS (Reference Images)

Popliteal artery branches & genicular anastomosis

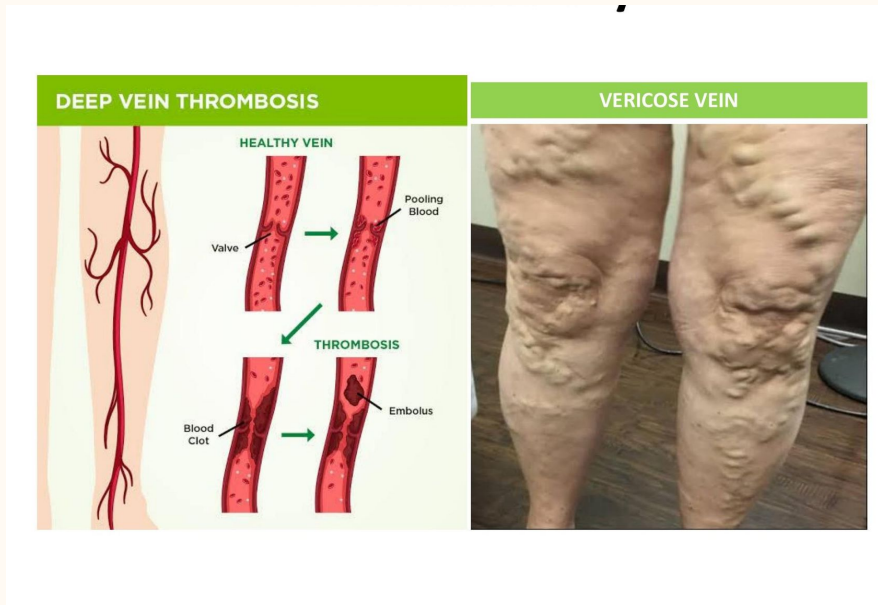


Genicular network detail + ankle arterial branches



Deep Vein Thrombosis & Varicose Veins

Alzheimer's Mode: A healthy vein valve is a one-way door — blood only goes up. Damage the door (varicose veins) and blood pools backward under gravity; block the pipe itself with a clot (DVT) and a piece can break off downstream as an embolus.



Deep Vein Thrombosis (DVT) — a blood clot forms within a deep vein (commonly the calf/popliteal/femoral veins), often from venous stasis, endothelial injury, or hypercoagulability (Virchow's triad). A fragment breaking free (embolus) can travel to the lungs, causing a life-threatening **pulmonary embolism**.

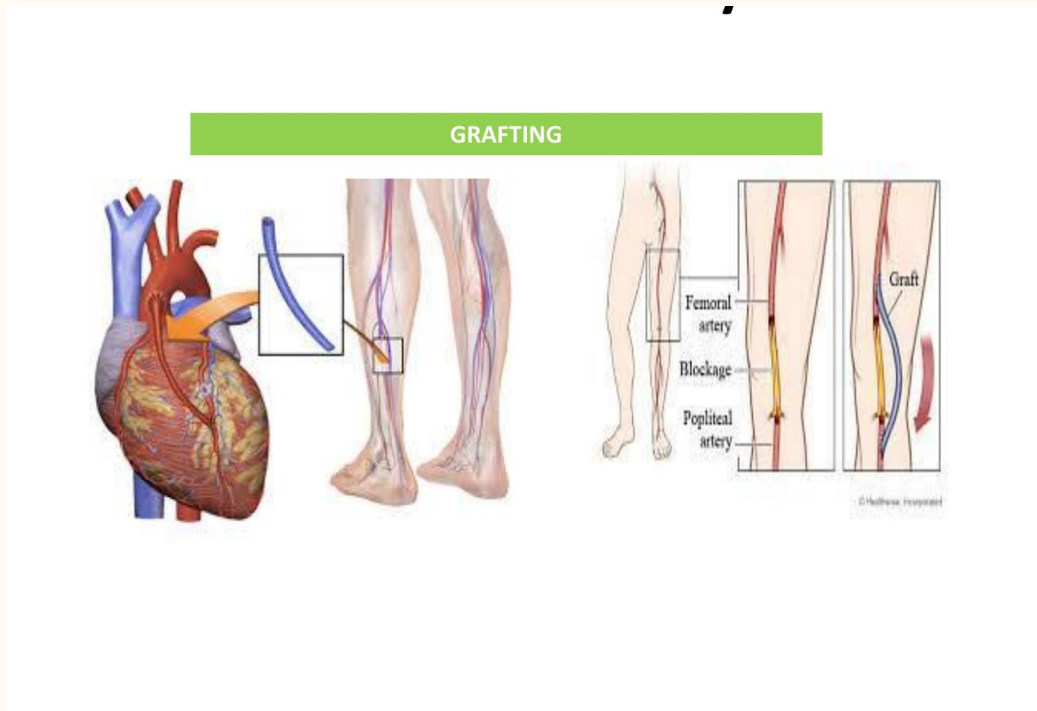
Varicose veins — superficial veins (great/small saphenous system) become dilated and tortuous when their valves fail, letting blood reflux and pool instead of returning against gravity. Risk factors: prolonged standing, pregnancy, obesity, family history.

EXAM HACK

DVT = deep system, clot, embolism risk. Varicose veins = superficial system, valve failure, reflux/pooling — not typically embolic. Both relate back to the same great/small saphenous → femoral/popliteal venous drainage pathway already covered under Veins.

Vascular Grafting — The Great Saphenous Vein as a Conduit

Alzheimer's Mode: The great saphenous vein is a spare length of "pipe" the body can afford to give up — surgeons harvest it and reroute it to bypass a blocked artery elsewhere, whether that's the heart or the leg itself.



Because it is long, superficial, expendable (extensive collateral/deep venous drainage takes over), and a good calibre match for arteries, the **great saphenous vein (GSV)** is the conduit of choice for:

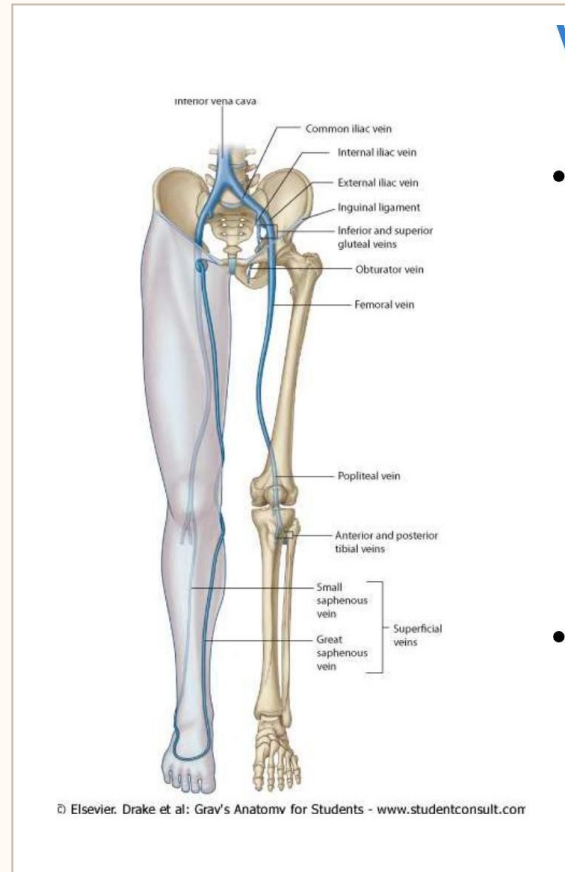
Coronary artery bypass graft (CABG) — a segment of GSV is harvested from the thigh/leg and grafted to bypass an occluded coronary artery.

Femoropopliteal bypass — used in peripheral arterial disease, where GSV is rerouted to bypass a blockage in the femoral or popliteal artery itself, restoring flow to the leg/foot.

CLINICAL

Harvesting the GSV is safe because the leg's venous return is preserved by the deep venous system and the small saphenous vein — the same redundancy that makes isolated great saphenous varicosities well tolerated.

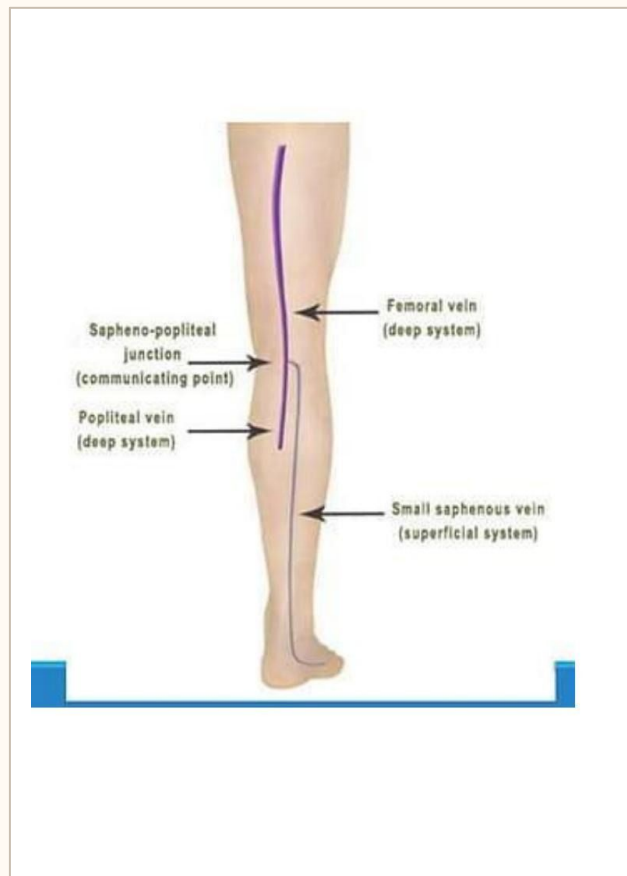
Deep veins (sharing names with arteries) and superficial great/small saphenous system



Reference image — supplementary visual aid, INDIGENE Study Console

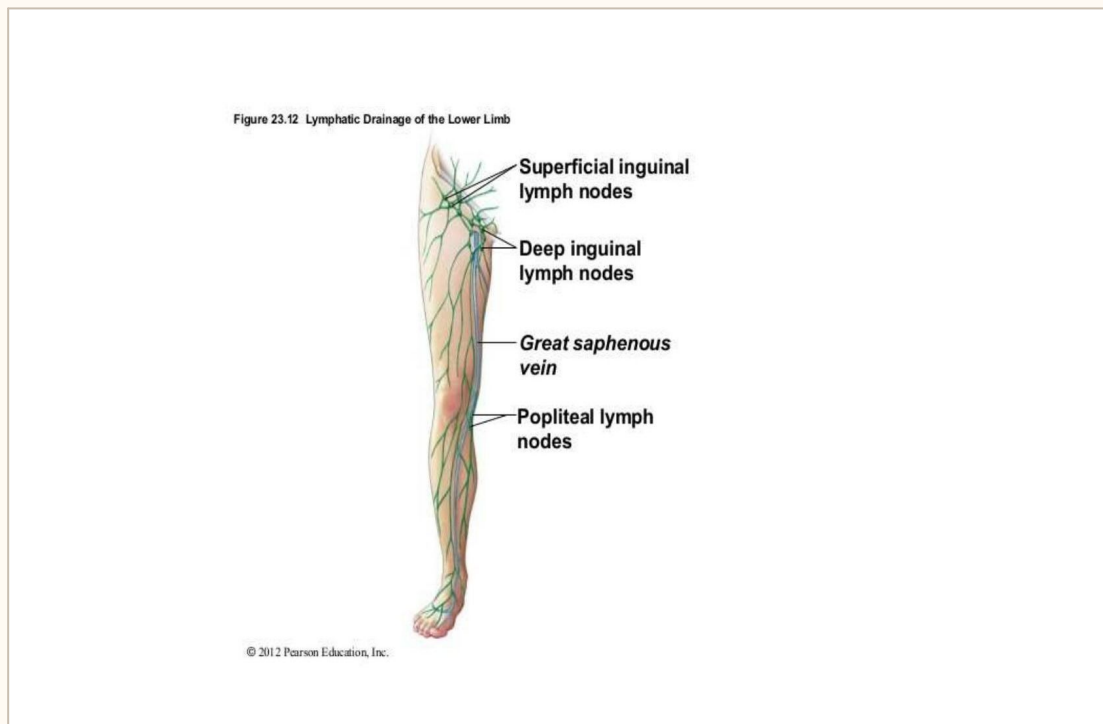
VASCULATURE — SMALL SAPHENOUS VEIN (Reference Images)

Small saphenous vein terminating at the sapheno-popliteal junction, 3–7.5cm above the knee



Reference image — supplementary visual aid, INDIGENE Study Console

Lymphatic drainage of the lower limb — superficial/deep inguinal and popliteal nodes



Reference image — supplementary visual aid, INDIGENE Study Console

6. FEMORAL TRIANGLE & POPLITEAL FOSSA

Femoral Triangle

Alzheimer's Mode: *The main doorway where the big neurovascular bundle transitions from the abdomen/pelvis into the leg.*

Boundaries: Superior = inguinal ligament (base). Lateral = medial border of sartorius. Medial = lateral border of adductor longus. Floor = iliopsoas + pectineus. Roof = fascia lata + superficial fascia.

Contents, lateral to medial — "NAVEL": Nerve (femoral) → Artery (femoral) → Vein (femoral) → Empty space (femoral canal, containing Cloquet's lymph node) → Lymphatics.

EXAM HACK

The femoral nerve lies OUTSIDE the femoral sheath — only the artery, vein, and canal are enclosed within it. The femoral canal is the site of femoral hernias, which appear below and lateral to the pubic tubercle (vs. inguinal hernias, above and medial) and carry a higher strangulation risk due to the canal's narrow, rigid boundaries. Femoral pulse is palpated at the mid-inguinal point (midpoint between ASIS and pubic symphysis — NOT the midpoint of the inguinal ligament).

The apex of the triangle (where sartorius crosses adductor longus) continues distally as the **adductor canal (Hunter's canal)**, carrying the femoral artery, femoral vein, and saphenous nerve to the adductor hiatus, where the femoral artery becomes the popliteal artery.

Popliteal Fossa

Alzheimer's Mode: *The knee's version of the cubital fossa — a diamond-shaped hollow where the big neurovascular bundle passes from thigh to leg.*

Boundaries: Superomedial = semitendinosus + semimembranosus. Superolateral = biceps femoris. Inferomedial/lateral = the two heads of gastrocnemius. Roof = popliteal fascia + skin. Floor = popliteal surface of femur, posterior knee capsule (reinforced by the oblique popliteal ligament), popliteus.

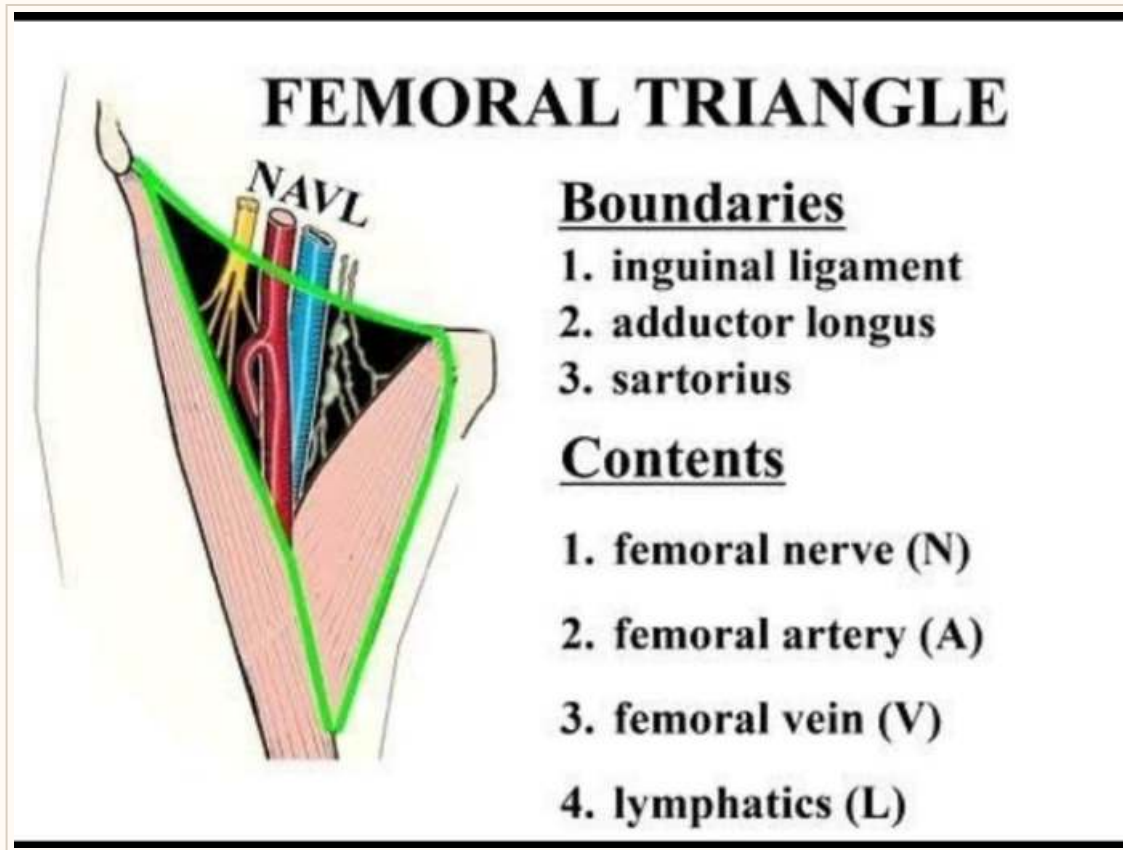
Contents, superficial to deep: Tibial nerve (most superficial, central) → Common peroneal nerve (along biceps femoris's medial border) → Popliteal vein → Popliteal artery (deepest, protected against the bone).

EXAM HACK

Depth order here is the REVERSE of the cubital fossa's "TAN" (Tendon-Artery-Nerve, artery in the middle) — in the popliteal fossa the artery is deepest, not the nerve. Small saphenous vein pierces the popliteal fascia to drain into the popliteal vein. Baker's cyst (popliteal cyst) forms in this space, often linked to underlying knee pathology. Posterior knee dislocation carries the highest risk of popliteal artery injury given its tethering above (adductor hiatus) and branching below.

FEMORAL TRIANGLE & POPLITEAL FOSSA (Reference Images)

Femoral triangle — boundaries & NAVL contents

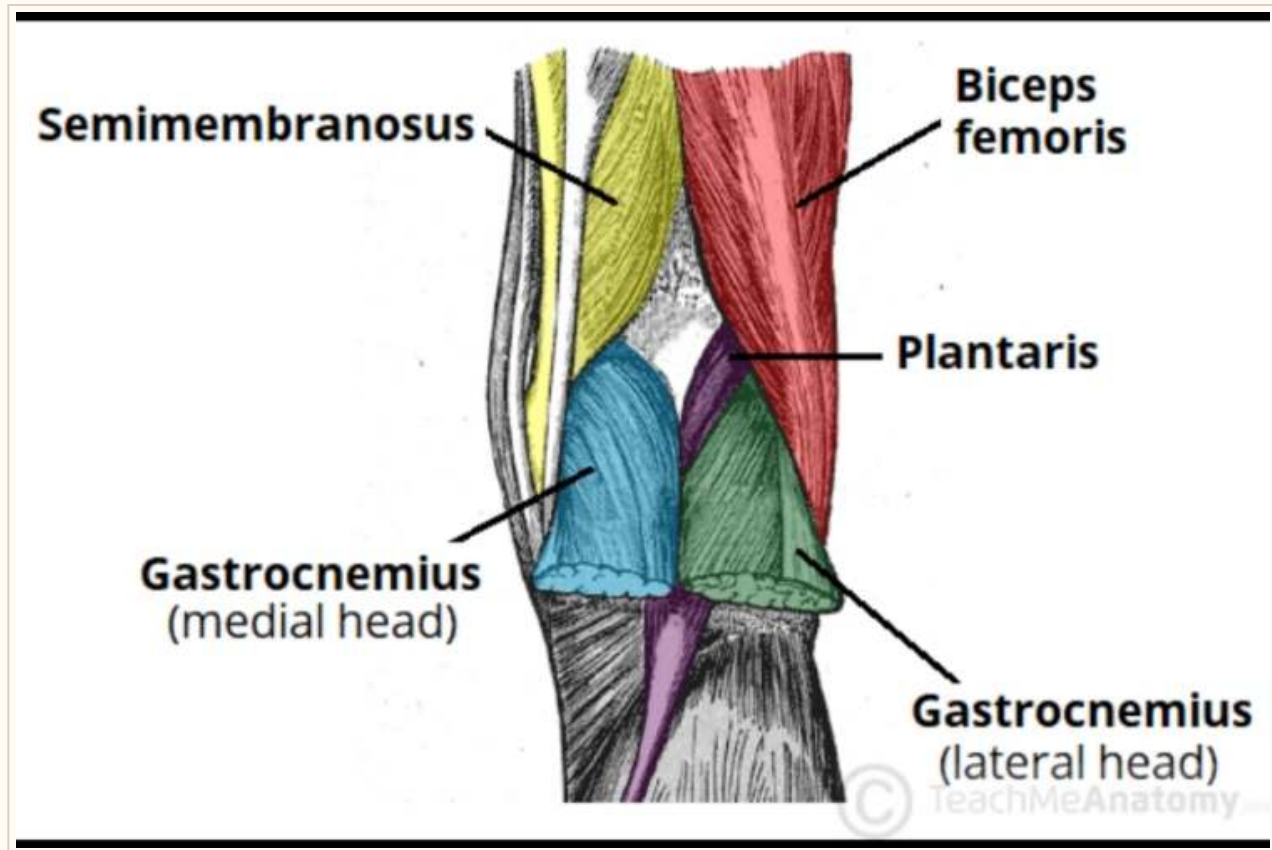


Popliteal fossa — boundaries, contents & floor

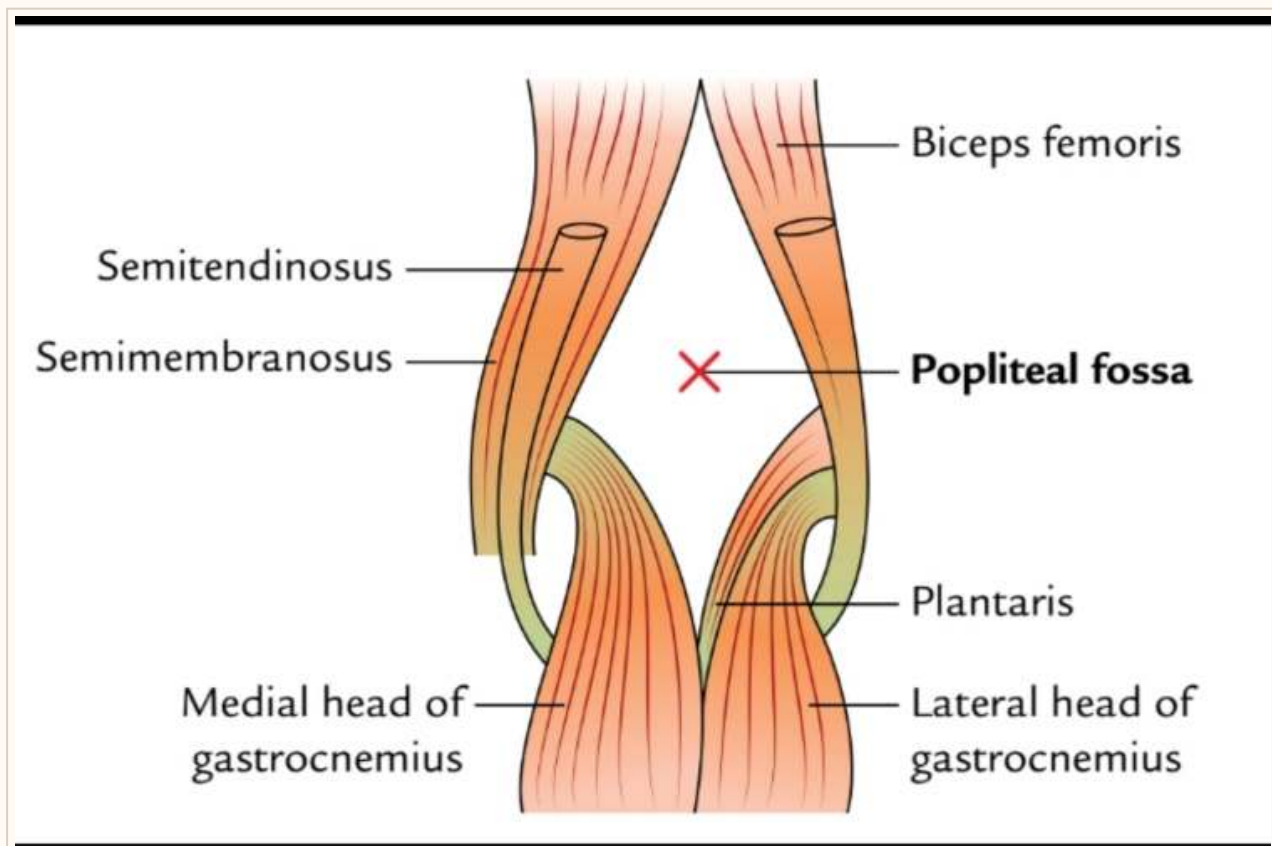
Popliteal Fossa					
Is a diamond-shaped intermuscular space at the back of the knee					
Boundaries				Contents	
Laterally	Medially	Roof	Floor	From medial to lateral	
(above) biceps femoris. (Below) Lateral head of gastrocnemius & plantaris	(above) semimembranosus & semitendinosus (Below) Medial head of gastrocnemius	1.Skin 2.superficial fascia & deep fascia of the thigh.	1.popliteal surface of femur 2.posterior ligament of knee joint 3.popliteus muscle.	1. Popliteal vessels (artery/vein) 2. Small saphenous vein 3. Tibial nerve 4. Common peroneal nerve. 5. Posterior cut. nerve of thigh 6. Connective tissue & popliteal lymph nodes. The deepest structure is popliteal artery.* (VERY IMPORTANT)	

POPLITEAL FOSSA — MUSCULAR BOUNDARIES (Reference Images)

Semimembranosus, biceps femoris, plantaris, gastrocnemius heads

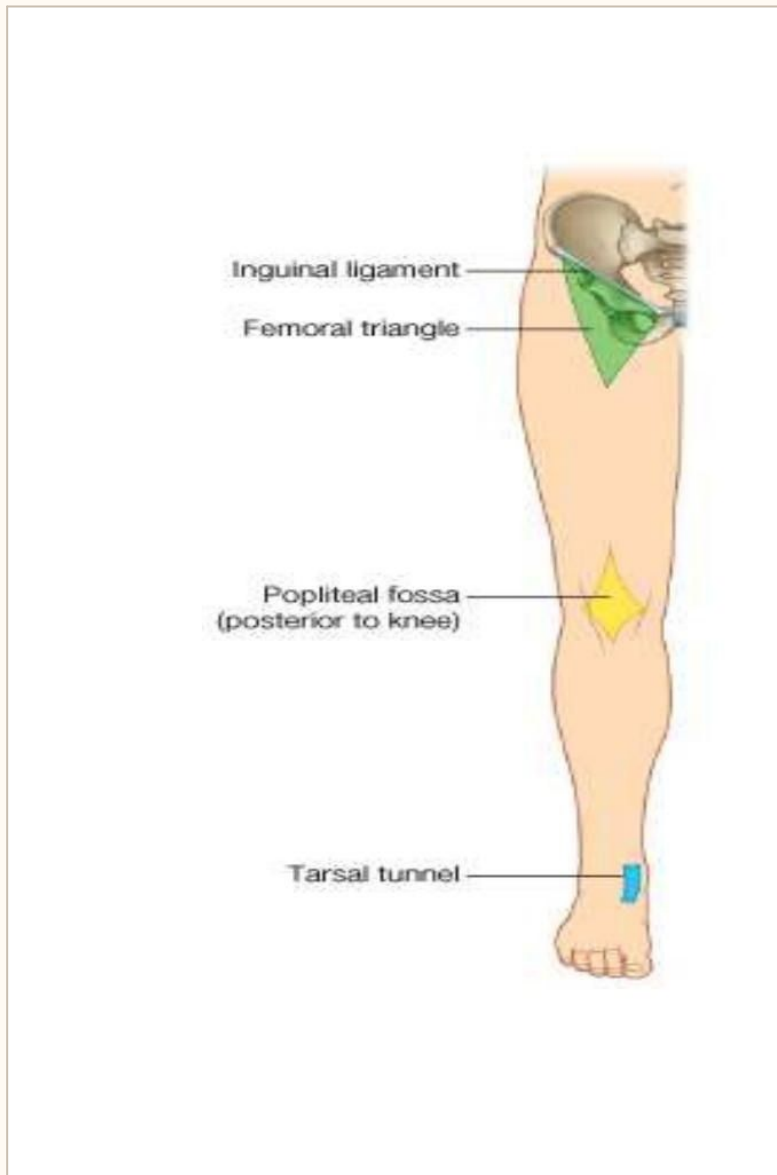


Popliteal fossa — diamond shape schematic



FEMORAL TRIANGLE & POPLITEAL FOSSA (Reference Images)

Femoral triangle, popliteal fossa, and tarsal tunnel — locations along the whole lower limb



Reference image — supplementary visual aid, INDIGENE Study Console

7. MASTER HACK SHEET — LAST-MINUTE REVIEW

Every mnemonic, trap, and high-yield rule from this compendium, compressed onto one section. Use this the night before the exam.

OSTEOLOGY

Hip bone: 3 bones (ilium/ischium/pubis) meet at the acetabulum. Tuberosity=hamstrings/sit; Spine=pudendal block landmark.
Femur: fovea capitis + foveal artery; quadrate tubercle=quadratus femoris; neck=most fracture-prone; screw-home mechanism from condyle asymmetry.
Tibia: intercondylar eminence w/ medial+lateral tubercles; fibular facet; soleal line. Fibula neck=common peroneal nerve.
Foot: "Tiger Cubs Need MILC." Talus=no muscle attachments. Sustentaculum tali on both talus & calcaneus.

JOINTS & LIGAMENTS

Hip: 3 extracapsular ligaments spiral, tighten in extension (Y-ligament=strongest in body).
Knee: 4 bursae (semimembranous, infrapatellar x2, prepatellar, suprapatellar). Transverse ligament links menisci. Oblique popliteal ligament=posterior capsule + popliteal fossa floor. MCL attached to medial meniscus; LCL not attached to lateral.
Ankle: Lateral=ATFL(weakest)/CFL/PTFL(strongest). Medial "deltoid"=4 named ligaments (tibionavicular, tibiocalcaneal, ant+post tibiotalar).
Sole ligaments: short plantar, long plantar (4 bands to metatarsals 1-4), spring ligament (calcaneonavicular, supports talar head+medial arch).

LUMBOSACRAL PLEXUS

Two linked plexuses: Lumbar (L1-L4, in psoas) + Sacral (L4-S4), joined by lumbosacral trunk.
Femoral n.=anterior thigh. Obturator n.=medial thigh. Sciatic n.=tibial+common peroneal bundled, splits at popliteal fossa.
Full cutaneous map: subcostal, genitofemoral (femoral branch), ilioinguinal, lateral/anterior cutaneous of thigh, obturator, posterior cutaneous of thigh cover the thigh; lateral cutaneous of leg, saphenous, superficial peroneal, sural cover the leg; superficial/deep peroneal, sural, tibial, medial calcaneal, medial/lateral plantar cover the foot.
Meralgia paresthetica=lateral cutaneous nerve of thigh entrapment.

GLUTEAL + THIGH MUSCLES

Superficial glutes=3 (max/medius/minimus+TFL); Deep=6 lateral rotators, 3 share one common tendon (obturator internus+both gemelli); quadratus femoris=own tendon+own nerve.
Medial thigh: pectineus=femoral n. (odd one out); adductor magnus=split (obturator+tibial n.).
Anterior thigh: psoas major (L1-L3 direct)+iliacus (femoral n.)=iliopsoas. Rectus femoris=only 2-joint quad head.
Posterior thigh (hamstrings): biceps femoris short head=common peroneal division (odd one out). Pes anserinus=sartorius+gracilis+semitendinosus, 3 nerves, "Say Grace before Tea."

LEG MUSCLES

Anterior compartment=100% deep peroneal nerve, dorsiflexors.
Lateral compartment=superficial peroneal nerve (except peroneus tertius=deep peroneal, it's really part of EDL).
Posterior compartment=100% tibial nerve. "Tom, Dick, AND Harry" behind medial malleolus=Tibialis post., FDL, Artery/Nerve, Hallucis longus flexor.
Gastrocnemius crosses knee+ankle; soleus crosses ankle only. Popliteus unlocks the knee.

FOOT INTRINSICS

Medial plantar n.="foot's median n." (smaller territory); Lateral plantar n.="foot's ulnar n." (majority).
Lumbricals: 2 medial=medial plantar n.; 2 lateral=lateral plantar n. (2:2 split, per your course material).
Quadratus plantae=no hand equivalent, corrects FDL's oblique pull.
Dorsal interossei have SPLIT innervation here (2 medial=deep peroneal, 2 lateral=lateral plantar) — unlike the hand's uniform ulnar supply.

VASCULATURE

Femoral a.: 5 early branches (superficial circumflex iliac, superficial epigastric, superficial+deep external pudendal, then profunda femoris).
Profunda femoris: medial+lateral circumflex femoral (+3 sub-branches each) + 4 perforating arteries. Cruciate anastomosis = med+lat circumflex femoral + sup+inf gluteal arteries.
Obturator artery: anterior branch (medial thigh) + posterior branch (acetabulum).
Genicular anastomosis: descending genicular + sup/inf medial+lateral genicular + circumflex fibular + recurrent anterior tibial = 6-7 vessel knee network.
Anterior tibial → dorsalis pedis = DORSUM. Posterior tibial → medial/lateral plantar = SOLE. Plantar arch = both systems meeting.

FEMORAL TRIANGLE & POPLITEAL FOSSA

Femoral triangle: Sup=inguinal ligament, Lat=sartorius, Med=adductor longus. NAVEL (lateral→medial): Nerve, Artery, Vein, Empty space, Lymphatics.
Femoral hernia=through the canal, below/lateral to pubic tubercle, high strangulation risk.
Popliteal fossa: Superomedial=semitendinosus+semimembranosus; Superolateral=biceps femoris; Inferior=gastrocnemius heads. Depth order (superficial→deep): tibial n., common peroneal n., popliteal vein, popliteal artery (REVERSE of cubital fossa's TAN).

RECURRING THEMES ACROSS THE LIMB

Retrograde blood supply → AVN risk: femoral neck, talus (mirrors scaphoid in the upper limb).
Split/dual innervation pattern: adductor magnus (obturator+tibial), biceps femoris short head (common peroneal only, not tibial), lumbricals 1 vs 2-4 (medial vs lateral plantar).
Screw-home mechanism (knee) mirrors the hip's ligament-spiral locking in extension — both reduce standing effort.
Fibular neck (common peroneal n.) parallels the humeral spiral groove (radial n.) — both superficial nerve-around-bone injury sites.

CLINICAL QUICK-FIRE

Femoral neck fx → shortened, externally rotated limb; AVN risk.
Posterior hip dislocation → most common, "dashboard injury," risks sciatic nerve.
Terrible triad (knee) = MCL+ACL+medial meniscus, from lateral blow to planted flexed knee.
Dashboard injury (knee, posterior force) → PCL tear.

Inversion ankle sprain → ATFL first, then CFL. Eversion → deltoid ligament strain or avulsion fracture.
Common peroneal n. injury (fibular neck) → foot drop + eversion weakness (both anterior+lateral compartments knocked out).
Tarsal tunnel syndrome = tibial n. compression behind medial malleolus, mirrors carpal tunnel.
Trendelenburg sign = superior gluteal n./abductor weakness → pelvis drops on the OPPOSITE side.

NEW ADDITIONS — QUICK-HACK ADDENDUM

Compressed mnemonics/traps for everything added in this complement pass — pairs with the Master Hack Sheet.

EXAM HACK

Duverney = isolated iliac wing fracture, pelvic ring intact, stable, conservative Rx.

EXAM HACK

Coxa vara <120° (Trendelenburg risk) · **normal** 120–135° · **coxa valga** >135° (DDH association).

EXAM HACK

Femoral fractures by site: **transcervical** = intracapsular, highest AVN risk · **intertrochanteric** = extracapsular, low AVN risk, heals well · **subtrochanteric** = extracapsular, high mechanical stress zone.

EXAM HACK

Crus fracture = tibial/fibular shaft break; tibia's subcutaneous border → often open; watch for compartment syndrome (6 P's).

EXAM HACK

Pipkin = femoral HEAD fracture + posterior hip dislocation. Type I/II by fragment position relative to fovea capitis; Type III adds neck fracture; Type IV adds acetabular fracture.

EXAM HACK

Sciatic nerve = tibial part from **ventral** divisions (L4–S3) + common peroneal part from **dorsal** divisions (L4–S2), one sheath, splits properly at the popliteal fossa. No direct cutaneous branches of its own.

EXAM HACK

DVT = deep system, clot, embolism/PE risk (Virchow's triad). **Varicose veins** = superficial system, valve failure, reflux/pooling, not embolic.

EXAM HACK

Grafting: great saphenous vein is the go-to conduit for CABG and femoropopliteal bypass — safe to harvest because deep venous system + small saphenous vein preserve leg drainage.